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Artificial intelligence in customer relationship management: A systematic framework for a successful integration

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ABSTRACT

This paper explores the integration of artificial intelligence (AI) applications into customer relationship management (CRM), focusing on how organizations can implement AI successfully in this context.

Using an explorative qualitative approach that includes interviews with AI experts, solution providers, and companies attempting AI-CRM integration, we identify key steps and practical activities often overlooked in literature and industry practice. Our findings provide a comprehensive framework for integrating AI into CRM, emphasizing aspects such as ethics by design, customer data centralization, model retraining, and ongoing user involvement.

This study offers academic contributions by extending theoretical insights on AI-CRM integration and provides practical guidance for executives and practitioners, helping align AI capabilities with CRM peculiarities to meet business needs.

1. Introduction

Customer relationship management (CRM) integrates marketing strategies with information technology (IT) to improve value for stakeholders by fostering meaningful relationships with key customers and segments (Lemon & Verhoef, 2016; Payne & Frow, 2005). Artificial intelligence (AI) – empowered by enhanced computing power, reduced costs, abundant big data, and advanced machine learning (ML) (Huang & Rust, 2021) – is viewed as the most desired emerging digital technology in revolutionizing CRM (De Bruyn et al., 2020). AI introduces new dimensions to CRM by enabling advanced analytics, automation, and personalized customer experiences (Kumar et al., 2020) throughout the customer relationship life cycle. AI can enhance customer acquisition by facilitating targeted marketing campaigns, improving conversion through predictive analytics and recommendation systems that optimize sales strategies, and strengthening loyalty by enhancing customer service with intelligent chatbots and self/robot-advisory solutions that provide real-time support. This technological integration requires careful consideration of data privacy, ethical concerns, and scalability, particularly in CRM settings in which data sensitivity and cross-functional coordination are crucial (Kaplan & Haenlein, 2019; van Esch & Stewart Black, 2021).

As indications of this disruption already are occurring, understanding remains limited as to how organizations can integrate AI into CRM successfully (Ledro et al., 2022), with a particular emphasis on identifying the main steps, activities, and actionable recommendations that characterize this integration process (Chatterjee et al., 2019; Davenport et al., 2020).

The extant literature has introduced some frameworks to help companies adopt and integrate AI applications into their operations, but research in the CRM context remains lacking, with its specific challenges and requirements (Herremans, 2021; Makarius et al., 2020; Reim et al., 2020). Unlike extant studies that focus on why organizations should adopt AI-CRM and what factors influence this decision (Chatterjee et al., 2019, 2021; Dastjerdi et al., 2023), we aim to investigate how to integrate an AI application into CRM. In so doing, we sought to answer the following research question: How does integration of an AI application into CRM take place?

Following an explorative qualitative approach based on several interviews with AI experts, providers, and companies that have attempted

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to integrate AI applications into CRM, we unveil the actions and steps involved in implementation and governance of AI in CRM – largely overlooked in both managerial practice and the literature.

As the AI-CRM integration literature remains in its infancy (Ledro et al., 2022), our explorative approach provides a systematic description of how AI-CRM projects have been implemented in practice. Building on the call for more empirical studies (Cao, 2021; Ozay et al., 2024), our findings offer detailed, case-based insights into AI-CRM projects' stages, sequences, and mechanisms, bridging the disconnect between AI technology and CRM strategy (Ledro et al., 2022). Thus, we provide a framework that highlights AI-CRM integration's unique and fundamental aspects, including establishing a culture of data awareness and attentiveness to customer needs from the outset, the need for ethics by design, customer data centralization, AI model retraining, and user involvement. This framework may serve as a reference for companies undergoing similar processes.

The paper begins by discussing the study's theoretical background in Section 2, focusing on integration of AI into CRM and highlighting its transformative impact and the need for tailored, strategic frameworks to align AI with CRM's unique aspects. Section 3 outlines the methodology, while Section 4 presents the results, detailing AI-CRM integration steps and key activities. Sections 5 and 6 discuss the findings and offer conclusions, including implications, limitations, and future research directions.

2. Theoretical background

2.1. AI's role in transforming CRM

CRM must be viewed as a strategic concept, rather than merely a technological one, that aims to enhance relationships through intelligent management systems (Saura et al., 2021). CRM components – such as operational CRM, analytical CRM, and collaborative CRM – provide structured approaches to managing different aspects of customer relationships, enabling a comprehensive understanding of customer interactions and increasing customer value (Alt & Puschmann, 2004; Rust & Huang, 2014).

Recent extant research has highlighted AI's transformative potential to enhance managerial and strategic processes within CRM (Kumar & Misra, 2021; Schröder et al., 2021), thereby identifying AI as the new frontier for CRM (De Bruyn et al., 2020; Mishra & Mukherjee, 2019). Indeed, AI significantly impacts all phases of the customer life cycle – acquisition, retention, and enhancement – transforming marketing, sales, customer service, and supply chain functions (Arco et al., 2019; Campbell et al., 2020; Kumar et al., 2020; Parvatiyar & Sheth, 2001). For example, in marketing, AI enables companies to predict market trends and customer behaviors, customize messaging for specific target audiences, and optimize advertising campaigns. In sales, AI helps identify best-selling product characteristics and improve sales strategies. In customer service, AI facilitates personalized customer support through human-like chatbots or guided assistance for service agents. In supply chain functions, AI helps forecast future events, such as product demand, as well as optimize supply chain operations.

2.2. AI integration into business operations

Implementing AI projects is challenging, with a high failure rate due to various critical factors, including organizational issues (complexity, poor planning, lack of top management support, cultural resistance), process-related challenges (ethical concerns, lack of expertise, and accountability), and technological difficulties (data quality, IT infrastructure, security, and integration complexities) (Bughin et al., 2018; Merhi, 2023; Mishra et al., 2022; Reim et al., 2020; Shrivastav, 2021). In light of these difficulties, extant research has proposed some step-by-step frameworks and models that might support companies in successfully integrating AI applications into business operations. Initially,

Table 1
Literature outlining steps for AI integration.

Study description	Steps for AI integration
Lee et al., 2019 Outcome: Steps for AI business model implementation Method: Literature review, case studies Addressed aspects: S; P	1 Execute pilot projects. 2 Build an in-house AI team. 3 Provide broad AI training. 4 Develop an AI strategy. 5 Develop internal and external communications.
Reim et al., 2020 Outcome: Steps for AI business model implementation Method: Systematic literature review Addressed aspects: S; P; D	1 Upper management involvement. Assess current and needed capabilities. Focus on data and infrastructure. Address transparency and risks. 2 Understand current BM and ecosystem role. Understand ecosystem position. Communicate findings to clarify AI usage. 3 Tailor capabilities to market needs. Focus on key capabilities. Manage transparency issues. Choose an implementation strategy. 4 Reach organizational acceptance. Mitigate resistance with pilot projects and training. Collaborate with partners. Implement feedback loops. Communicate to minimize uncertainty.
Makarius et al., 2020 Outcome: Steps for bringing AI into the organization Method: Conceptual Addressed aspects: S; P	1 Sensing: Communicate efforts to employees, provide resources, and offer training. 2 Comprehending: Clarify roles, determine task mastery, social interactions. 3 Acting: Build an understanding of how AI system will interact with human users and observers. Figure out role transitions and trust development. 4 Employee learning.
Herremans, 2021 Outcome: Steps for developing a successful AI strategy Method: Conceptual Addressed aspects: S; P; D	1 AI project goals and problem statements. Involve team with domain and AI expertise. Select top projects for further evaluation. 2 Assess data availability and quality. Decide on data collection vs. acquisition. Address legal issues. 3 AI Team Strategy: Establish an interdisciplinary AI team. 4 Determine team positioning within the organization. 5 Address challenges like bias and model legality. Keep humans in the loop. 6 Define value-based key performance indicators for project tracking. 7 wt risks and benefits. Classify shortlisted projects based on risk strategy. 8 Promote a culture of adoption. Provide ongoing training for employees.
Shrivastav, 2021 Outcome: Steps of AI lifecycle in Supply Chain Management Method: Conceptual Addressed aspects: T; S; D	Steps: 1 Problem identification. 2 Data preparation. 3 Training and validation. 4 Scaling and employment. 5 Feedback and automation.
Fenwick et al., 2023 Outcome: Steps toward human-centric AI in HR Method: Conceptual Addressed aspects: P	Steps: 1 Initial integration of AI automates HR functions like planning, recruitment, training, and performance management. 2 AI and humans work closely, integrating AI into daily functions and personalizing employee experiences. 3 Mature stage where human resource management manages human-AI interaction to enhance overall human experience and create a technology-driven workplace for societal benefit.
Bonetti et al., 2023 Outcome: Steps for AI co-evolution in Retail Method: Interviews, on-site observations Addressed aspects: P	1 Co-envisioning. 2 Co-adapting. 3 Co–(re) aligning.
Holmlund et al., 2020 Outcome: Steps for how to use BDA for CX management Method: Conceptual Addressed aspects: S; P; D	1 Determine CX actions. Start small. Obtain necessary resources. 2 Evaluate CX insights and analytics needed. Consider in-house resources or outsourcing. Determine analytics comfort level. 3 Analyze CX data availability and ownership. Assess storage and integration capabilities. 4 Decide on analytics and data strategy. 5 Involve internal and external stakeholders. Utilize modern user experience

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Table 1 (*continued*)

Study description	Steps for AI integration
	approaches. Scale solution based on organizational needs. Ensure reliability and continuous monitoring. 6 Establish feedback loops. Share insights across the organization. Create cross-functional teams. Measure impact on business metrics.

Note: T: Technological Development; S: Strategy formulation and culture cultivation; P: People engagement; D: Data management.

within the realm of information systems, many studies have put forth software development life-cycle (SDLC) models (e.g., Boehm, 2007; Ebert et al., 2016; Mathur & Malik, 2010; Royce, 1987), which provide practitioners with several steps that should be followed from a technical perspective to develop and implement software within their organizations successfully. As the AI wave accelerates, authors of extant literature have shifted attention toward the process of integrating AI into business operations, proposing some models for AI development life cycles and AI integration that differ from traditional SDLC models due to several factors: AI systems are probabilistic and experimental; require significant data and resources (Laato et al., 2024); and continue to evolve based on feedback loops and increasing data availability (Shrivastav, 2021). These complexities, along with biases and ethical concerns, necessitate specialized strategies for AI integration (Birkstedt et al., 2023). As presented in Table 1, AI integration and development lifecycle models present the steps that managers should follow to adopt AI, ranging from design (or predevelopment) to deployment (or scaling) (De Silva & Alahakoon, 2022; Lee et al., 2019; Makarius et al., 2020; Reim et al., 2020). These studies have provided a diverse array of AI integration, placing greater emphasis on aspects of strategic formulation and culture cultivation (Herremans, 2021; Lee et al., 2019; Reim et al., 2020), people engagement (Bonetti et al., 2023; Fenwick et al., 2023; Makarius et al., 2020), and data management (Cao, 2021). They also have outlined linear steps toward AI integration, including defining strategic goals, building interdisciplinary teams, preparing data, and maintaining transparency.

Also, practitioners (Daugherty et al., 2024) and consulting companies, such as Accenture (Kanioura & Lucini, 2020) and Capgemini (Stancombe et al., 2017), have contributed actively to development of guidelines and models, emphasizing the importance of transparency, privacy protection, and stakeholder engagement. Notably, such models are not prescriptive, but instead provide a strategic framework for introducing AI into an organization without delving into detailed steps or specific contexts.

As Table 1 indicates, some studies have focused on integrating AI into specific contexts, such as customer experience (CX) management (Holmlund et al., 2020), supply chain management (Shrivastav, 2021), human resource management (Fenwick et al., 2023) and retail (Bonetti et al., 2023; Cao, 2021). Jöhnk et al. (2021) emphasized the need for context-specific and purpose-specific considerations when integrating AI. In addition, most of the articles that have examined AI integration are conceptual or based on secondary data, leaving room for collecting primary data to explore and deepen our understanding of the subject further (Cao, 2021).

2.3. AI integration into CRM

The extant literature in the field of AI in CRM mainly has focused on why organizations should adopt AI and what factors influence this decision, rather than how to integrate and scale AI. For example, Chatterjee et al. (2019) first developed a conceptual framework to check whether an organization is ready to adopt AI for CRM; they later proposed a theoretical model grounded in Technology Acceptance Model 2, highlighting the importance of organizational agility, social involvement, and employee trust in adoption (Chatterjee et al., 2021). Subsequently, Dastjerdi et al. (2023) expanded on this by incorporating external pressures, such as policy and customer demands, emphasizing the need for adaptability, particularly in sectors such as healthcare. While these studies mainly have focused on factors that push firms to adopt AI in CRM, very little extant literature has examined the process of integrating AI into CRM, as Ledro et al. (2022) highlighted. Only Holmlund et al. (2020) conceptualized a step-by-step guide for managers on "how to use big data analytics (BDA) for CX management" (see Table 1).

Therefore, empirical research based on primary data is needed to deepen understanding of how AI can be integrated successfully into CRM, especially given such contexts' complexity.

Indeed, CRM strategies have specific peculiarities that should be considered when planning and implementing AI integration into CRM. First, they involve integrating data from disparate sources (e.g., marketing automation tools, social media platforms, customer support systems, or e-commerce platforms) (Rust & Huang, 2014; Saarijarvi et al., 2013), often depending on sensitive customer information, thereby raising concerns about privacy, security, and multi-channel integration (Kumar et al., 2020). Efficient gathering, utilization, and analysis of customer data are crucial to implementing CRM strategies successfully (Payne & Frow, 2005). Along the same lines, CRM has raised ethical considerations – particularly in terms of data use, biases, and transparency (van Esch & Stewart Black, 2021) – that need to be addressed through ethical guidelines, including monitoring and auditing mechanisms to ensure fairness and accountability in CRM operations (Kaplan & Haenlein, 2019).

Second, CRM strategy relies on customer behavior, trends, and preferences to provide customization options, which may introduce complexity in configuration, data mapping, synchronization, and real-time updates (Graesch et al., 2021; Rust & Huang, 2014). Additionally, CRM's success depends on a range of organizational capabilities – such as clear communication, leadership support, knowledge management, and adaptability to changes – all of which must be considered during AI integration to align organizational efforts effectively (Cambra-Fierro et al., 2017; Croteau & Li, 2003; King & Burgess, 2008).

Third, CRM's inherently cross-functional nature – encompassing multiple departments beyond just marketing, sales, and service – demands collaboration across diverse organizational functions, such as product development, finance, and operations (Alt & Puschmann, 2004). However, siloed information structures, which are common in CRM environments, often obstruct communication and coordination, creating challenges in effective integration (King & Burgess, 2008; Malthouse et al., 2013). Therefore, collaborative complexity must be addressed thoroughly during AI integration to ensure effective alignment across all teams.

Finally, as the customer base grows and data volume increases, scalability and efficient workload management become key considerations in CRM strategies (Perna & Baraldi, 2014), which should be properly considered when introducing an AI application.

Overall, studies tailored to AI integration into CRM are necessary to ensure that AI integration efforts are aligned with CRM strategies and processes' unique needs.

In the face of these proven difficulties, deeper research and empirical evidence are needed on how companies can integrate AI applications into CRM successfully, leaving room for research that further explores the necessary steps, drivers, enablers, barriers, and challenges tied to a successful AI application in CRM (Chatterjee et al., 2019; Davenport et al., 2020; Loureiro et al., 2021). In this regard, proposing a framework to help managers successfully integrate AI into CRM is crucial. It helps address potential failures by recognizing the necessary expertise and breaking down complex concepts into manageable steps.

3. Research methodology

We followed an inductive qualitative approach (Gioia et al., 2013),

Table 2
Sample characteristics.

ID	Main industry	Size	ISO Country Code	Level of success	Goal of AI integration into CRM	Interviewee ID
ROUND 0						
C0	Tech	50–200	IT	S	Enhance customer engagement by prioritizing high-value clients and accurately interpreting their needs for timely and personalized service	AI/ML-CRM project manager – C00
ROUND 1						
CA	Plastic and rubber manufacturing	1–50	IT	S	Improve decision-making and customer engagement by prioritizing high-value clients through advanced email and text analysis	AI-CRM manager – CA1
CB	Corporate & Investment Banking	10k-50k	FR	S	Enhance customer retention and satisfaction by predicting detractors, managing risks, and understanding customer needs for proactive interventions	Head of R&I – CB2; Director of innovation & CX – CB3
CC	Insurance	>50k	CH	U	Streamline operational efficiency by automating claims processing and document management	Head of project, insurance manager – CC4
ROUND 2						
CD	Foodservice equipment manufacturing	200-1000	IT	U	Improve efficiency in customer management by prioritizing high-value opportunities and automating document handling processes	Head of project, IT manager – CD5
CE	Insurance	10k-50k	IT	S	Proactively manage customer relationships by identifying at-risk clients and mitigating potential issues	Lead data scientist – CE6
CF	Food retail	>50k	FR	S	Enhance customer engagement and retention by utilizing AI for personalized marketing strategies, effective segmentation, and proactive intervention for at-risk customers	Group data lead – CF7
PA	Tech	10k-50k	FR (HQ USA)	–	Provider of software and technologies for databases. Through its AI-driven applications, it aims to empower businesses with advanced analytics, automation, and CRM tools, facilitating better decision-making and improving overall CXs across various industries	CX specialist – PA8
PB	Tech	1–50	FR	–	Consultancy that supports AI transformations and helps unlock AI's potential for seamless integration, tailored solutions, and streamlined processes for clients	Founder & independent advisor – PB9
PC	Tech	50-200	FR	–	B2B independent software vendor developing AI to empower insurance and bank insurance players fostering long term relationships with customers	Head of marketing & communication – PC10
PD	Tech	1–50	FR	–	Provided of AI/ML solutions that aim to facilitate the daily life of sales teams and their managers	Chief executive officer – PD11
PE	Tech	1–50	FR	–	Consultancy that provides accurate vision of AI winning strategies devoted to better understand customer needs, provide a better customer service, and identify best offers for customers	Associate partner – PE12
ROUND 3						
CG	Telco	>50k	IT	S	Enhance CX through 24/7 support via a virtual assistant, personalized online recommendations, social media analysis, sales optimization, and automation through chatbots to streamline operations and improve efficiency	AI center of excellence director – CG13 Innovation project manager –CG14
CH	Food retail	10k-50k	IT (HQ FR)	S	Deliver a seamless and personalized CX across all channels while effectively managing risks and proactively addressing customer needs to enhance retention and satisfaction	Chief information officer – CH15; Chief data officer – CH16
CI	Automotive dealership	50-200	IT	U	Assess relationship intensity and prioritizing high-value opportunities to maximize impact and improve overall customer satisfaction	CRM manager – CI17
PF	Tech	200-1k	IT	–	Developer and provider of solutions to improve the interaction between companies and their customers, using AI and advanced language understanding technologies	Chief technical officer – PF18
EA	–	–	FR	–		Expert in digital & AI in marketing – EA19
EB	–	–	IT	–		Expert in AI & business development – EB20
EC	–	–	NO	–		Expert in machine ethics –EC21
ROUND 4						
CJ	Insurance	>50k	FR	S	Proactively improve CX and retention by predicting detractor behavior, managing risks, personalizing offers, and utilizing advanced analytics for effective communication and documentation processes	Chief data officer –CJ22
CK	Fashion Retail	50-200	IT	S	Improve and personalize the CX along the customer journey	Chief digital officer –CK23
CL	Banking and financial services	>50k	FR	S	Enhance customer engagement and retention through effective segmentation, personalized offer recommendations, automated documentation processes, and proactive intervention for potential detractors	Chief executive officer –CL24
CM	Consumer finance	1k-5k	FR	S	Better know and interpret customers' needs, and improve customer service across all channels	Digital and development project manager – CM25

Note: **ID**: first letter = sample group (C as company that tries to integrate one or more AI applications in CRM, P as provider/consultant of AI applications in CRM, E as expert on specific topics with more than 20 years of experience); second letter = number of the interviews within the group. **Level of success of AI integration**: S=Success; U=Unsuccess

Table 3
AI applications along CRM processes of our interviewed companies.

CRM process	AI application
Marketing	Predictive Analytics for customer segmentation (CF; CL) Natural language processing (NLP) and automation to automate and customize email/promotion (CF) Predictive Analytics to calculate indirect effect of promotions (CF) Recommendation System to create a catalogue auto filled and more personalized (CF; CH) Recommendation System to personalize value proposition online (suggestions and recommendations) (CF; CG; CH) Sentiment analysis of conversations for follow up prioritization (CM) Sentiment Analysis and Social Listening using NLP for analysis of online word of mouth, tweets (CG) Predictive Analytics to identify and optimize the level of relationship intensity with potential customers or customers (C0; CI) Predictive Analytics for predicting potential detractor customers and targeting them for intervention or mitigation efforts (CB; CE; CF; CH; CJ) Machine learning to optimize the omnichannel strategy (CH; CM)
Sales	Supervised machine learning to understand and manage risks (pricing, fraud detection) (CB; CE; CH; CJ) Predictive Analytics to prioritize clients/opportunities based on their value/urgency (CA; CD; CI) Recommendation System to automatically identify the best offer to recommend to customers (CJ; CL) Recommendation System to better identify potential offers and make recommendations to sales team (CG; CJ)
Customer service / After sales	Recommendation Systems and Personalization Algorithms for delivering the same CX and recommendation online and offline (CH) Personalization Algorithms for improving and personalizing the CX along the customer journey (CJ; CK) Chatbot/conversational agent (CG; CM) Email and text analysis and classification with deep learning-based NLP (CA; CC; CJ; CM) Automatic claims management (CC) Automatic reading, classification, and transfer of documentation (CC; CD; CJ; CL) Predictive Analytics to better know and interpret customers' needs to better serve them / predict life events (C0; CB; CH; CM) Recommendation Systems and machine learning to optimize the assortment management to better serve customers (CF; CH)

Note: ID in the bracket refers to the interviewed company, see Table 2.

which is particularly suitable for exploratory research (Saunders et al., 2019) and useful for gathering explanations from the involved stakeholders (Patton, 2002). By taking a multiple case study approach, we gained comprehensive insights by comparing various companies across different industries and analyzing empirical data from diverse, information-rich primary sources (Eisenhardt & Graebner, 2007). This method also provides reliable insights for future research and a deeper understanding of unexplored topics (Boddy, 2016). Primary data were collected through semi-structured interviews.

Data collection and analysis proceeded in tandem, and ongoing analysis steered the course of inquiry (for a similar approach, see Ritala et al., 2024). Example questions were as follows: Can you describe how your project to integrate AI into CRM originated and how it evolved? What steps and people were involved from the genesis of the project to its use? What difficulties and problems did you encounter before, during, and after implementation, and how did you solve them? The questions were formulated in a nondirectional and disconnected way to avoid active listening (interviewer-induced bias) (Bryman et al., 1990), and we strived to clarify ambiguities.

3.1. Sample and data collection

We applied purposeful sampling with maximum variation to follow a replication logic (Yin, 2014), starting with selective sampling, which allowed us to develop the conceptual lines that guided our theoretical sampling (Coyne, 2018), a nonrandom sampling method aimed at building concepts from data (Coyne, 2018), with an emphasis on including participants who offer deeper insights into the topic (Homburg et al., 2017). Selective sampling and theoretical sampling are both forms of purposive sampling (Coyne, 2018). Due to our subject's novelty, we used different channels to recruit participants (e.g., we participated in conferences on AI and big data that focused on CRM, used our network of contacts, and read articles in specialized magazines and journals).

Our interviewees had at least five years of experience in CRM or related operations, held top management C-level positions, and were directly involved in managing integration of one or more AI applications in CRM. To reduce memory issues and retrospective bias among our informants (Eisenhardt & Graebner, 2007), we interviewed those who had taken part in AI integration into CRM within a maximum time frame of one year since the project's launch.

The interviews occurred between June 2021 and May 2022. Throughout each round of data collection and analysis, emerging concepts were examined by incorporating additional interview questions for subsequent informants, who were selected to validate or contest the perspectives gathered so far (Gasson, 2009). We conducted four rounds of data collection and analysis comprising 25 interviews (see Table 2). After Round 0, during which we tested the protocol with a tech company that both internally integrates and sells an AI application, we started with a selective sampling of maximum variation (Coyne, 2018), involving companies that have demonstrated different experiences with AI integration into CRM – successful or unsuccessful – and that belong to different industries (see Table 2). AI integration into CRM is deemed successful if "participants indicated that the system (intended as AI integration into CRM) was satisfactory in meeting its objectives" (Chen et al., 2009). Based on the insights from these interviews, we moved to the second round, to which theoretical sampling was applied. Indeed, the first round demonstrated the need to expand the first group of samples (companies) to the retail sector to cover more applications of AI in CRM and simultaneously expand our work into a second group of samples (providers or consultants of AI applications in CRM) to gain a broader perspective and more precise explanations (see Table 2). The AI applications considered, as presented in Table 3, cover a variety of CRM processes (marketing, sales, and service).

The insights from the second round pushed us to expand our work by interviewing more companies, and a third group of samples (experts) was used to cover specific topics, such as ethics and human–machine relationships. At the end of the third round, we achieved approximate theoretical saturation. During the fourth round, no new insights emerged, i.e., theoretical saturation was reached (Boddy, 2016; Guest et al., 2006). The sample size fell within an appropriate range for achieving maximum variation (Kuzel, 1992).

In line with extant research (Gupta & Ramachandran, 2021; Malodia et al., 2020), we followed a data triangulation approach. During interviews, respondents referenced relevant documents, while parallel searches of public databases were conducted to ensure data completeness and precision. Secondary sources included company websites, official documents, news media articles, public videos, interviews, and reports. In cases when this was not possible, we consulted a second interviewee (Perry, 1998).

3.2. Data analysis

We analyzed all data with the support of ATLAS.ti software and followed Strauss and Corbin's (2000) coding procedure. We used an open-coding procedure to identify preliminary concepts, then conducted

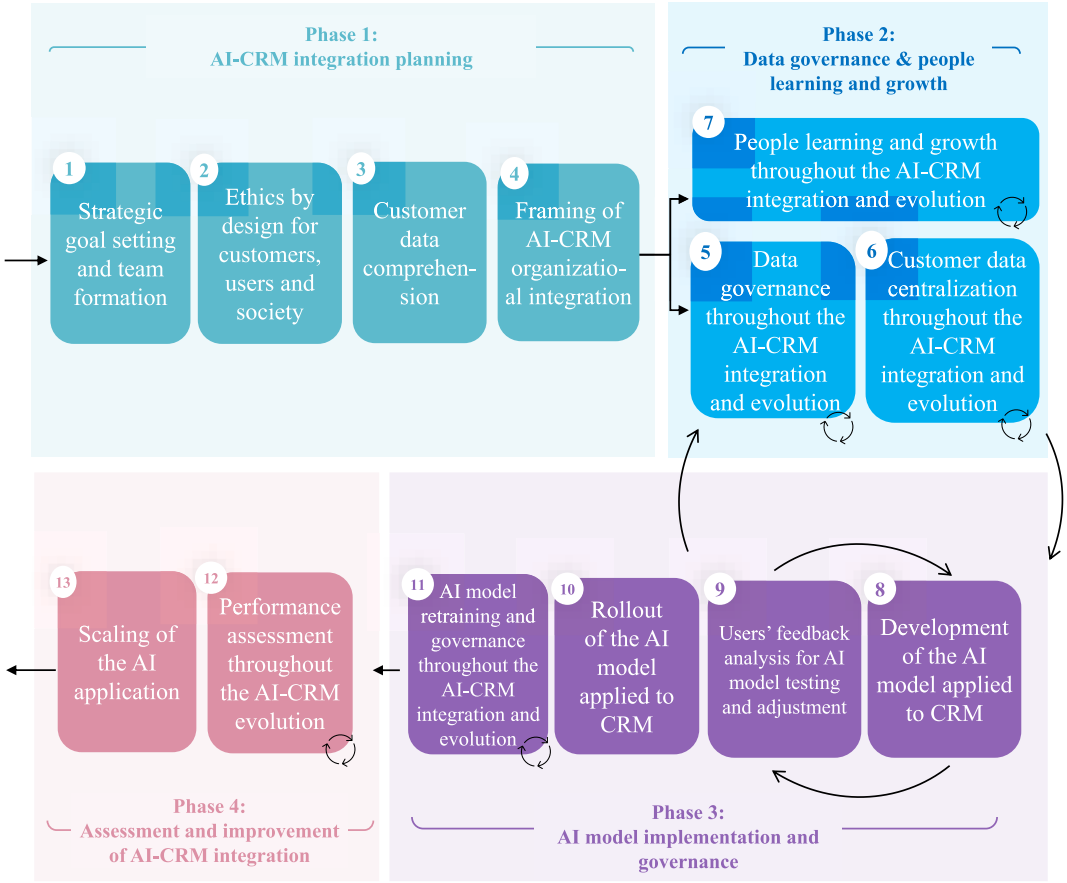

Fig. 1. Framework of the integration of a successful AI application in CRM.

axial coding to group concepts, iteratively based on the same notion, into first-order categories (Gioia et al., 2013). To provide a higher level of abstraction and organization, we then grouped these first-order categories into second-order categories (Gioia et al., 2013). Finally, we continued to assemble them at a more abstract level to identify the aggregate core categories. All this took place iteratively, with constant shifts back and forth between codes and data, and with emerging ideas leading to additional data collection as the research progressed (Langley & Abdallah, 2011). Furthermore, we constantly compared emergent concepts with extant literature (Eisenhardt, 1989).

First, we conducted a within-case analysis to identify the pattern that participants have followed to integrate AI into CRM (Yin, 2014), followed by a cross-case comparison of successful and unsuccessful cases (Langley, 1999; Puliga et al., 2023). This approach allowed us to capture patterns across different contexts and industries. Despite variations in AI application types, the cross-case analysis highlights common integration strategies across cases. Additionally, insights from consultants and experts with experience across multiple AI-CRM implementations provided further validation, reinforcing the existence of a unified integration process applicable to diverse AI-CRM projects. Based on the results from this cross-analysis (Yin, 2014), we developed a systematic framework for integration of an AI application into CRM and a question guide for managers.

To validate our findings further, we consulted experts from Salesforce and Microsoft Dynamics (see Appendix B).

4. Results

The open coding process began with identification of key quotes from the interview data. These quotes served as the foundation for the development of first-order categories, which were reduced from more than 100 initial first-order categories to a manageable 46 appropriate first-order categories. The application of axial coding resulted in 13 s-order categories and four aggregate dimensions in the data structure. The four aggregate dimensions corresponded to the framework's macro-phases (see Fig. 1).

For each macro-phase, the specific steps (which corresponded to the 13 s-order categories) and key activities (which corresponded to the 46 first-order categories) were identified (see Table 4). The AI-CRM integration steps, while presented linearly, are often iterative, requiring revisits and adjustments throughout the process. The focus should be on effective execution of these steps, rather than their sequence. As presented in Fig. 1, Steps 5 to 9 must not be interpreted as a linear process, as some loops can occur. For example, Step 7 can start gradually, beginning with a brief introduction to the tool, and then progress into structured training when Step 9 is reached.

Appendix A (Table A1) lists participants' quotes that illustrate how to implement each activity in the AI-CRM integration process. Each quote is attributed to a specific interviewee, referenced by an alphanumeric ID (see Table 2). These quotes are linked directly to the categories that emerged during the open coding process and ultimately are connected to the higher-level themes discussed in this section

4.1. Macro-phase one: AI-CRM integration planning

This initial phase ensures that AI integration into CRM aligns with corporate goals, ethical principles, data availability, and privacy concerns.

Step 1: Strategic goal setting and team formation. Participants stressed the importance of formulating, from the very beginning, the business goals expected from the AI application and related performance measures, defining specific key performance indicators (KPIs) (e.g., CB2;

Table 4
Coding results and category development.

First-order categories	Second-order categories	Aggregate dimensions
– Formulate clear business goals. – Clarify the performance you want to achieve – Multidisciplinary team formation	1. Strategic goal setting and team formation	1. AI-CRM integration planning
– Establish an ethics committee – Discuss and define ethical principles – Involve stakeholders – Be collaborative and responsible	2. Ethics by design for customers, users and society	
– Data capital comprehension – Data usefulness comprehension – Data access comprehension	3. Customer data comprehension	
– Clarify how to achieve your goal through AI. – Decide whether to make or buy the AI solution – Decide ownership, sponsor, people involved, and deadline ambitions – Analyze the external environment – Build a multidisciplinary, multicultural, and multiskills team – Plan how to ensure ethics principles – Set output and boundaries – Plan how to measure and read the results – Plan how to embed the model in business operations – Plan how to scale the AI application	4. Framing of AI-CRM organizational integration	
– Data modelling and creation – Data collection – Data validation – Validation of the AI ethics principles of data privacy and regulation, fairness, and well-being	5. Data governance throughout the AI-CRM integration and evolution	2. Data governance & people learning and growth
– Customer data structuring – Customer information reorganization – Gather and consolidate all customer-related information into a single centralized system	6. Customer data centralization throughout the AI-CRM integration and evolution	
– Training of all people to stimulate trust and awareness around AI – Validation of the AI ethics principles of accountability and well-being	7. People learning and growth throughout the AI-CRM integration and evolution	
First-order categories	**Second-order categories**	**Aggregate dimensions**
– AI model training – Validation of the AI model – Validation of the AI ethics principles of explainability and transparency	8. Development of the AI model applied to CRM	3. AI model implementation and governance
– AI model testing – Achieve a tradeoff between performance and ethical issues – Users' feedback analysis before roll-out – Humans on the loop to take corrective actions – Data and concept drift monitoring – Validation of the AI ethics principles of fairness (prevent discrimination)	9. Users' feedback analysis for AI model testing and adjustment	
– Integrate the AI model within the IT system – Rollout of the final model	10. Rollout of the AI model applied to CRM	
– AI model retraining – Users' feedback analysis after roll-out – Monitoring and correction of AI model, its inputs and outputs	11. AI model retraining and governance throughout the AI-CRM integration and evolution	
– Measure and monitor key performance indicators – Check and adjustment of the AI model and its performance assessment	12. Performance assessment throughout the AI-CRM evolution	4. Assessment and improvement of AI-CRM integration
– Scale of the entire portfolio, business, or all geographies	13. Scaling of the AI application	

CF7; CM25; PE17; see Table A1 for full quotes). These KPIs fall into five key categories: algorithm performance; customer metrics; organizational sustainability; processes; and user/employee impact (see Table A3 for more details). When the broad objective is broken down into key goals, the likelihood of success in AI-CRM integration increases (CB2; PA8). However, companies that started with very ambitious and challenging goals failed to implement AI (CC4). An effective strategy involves "taking a top-down approach: This includes the maturity analysis, impact analysis, and the matrix to select key priorities" (CF7).

A top-down strategy ensures alignment with organizational goals and provides clear direction for AI-CRM integration efforts, guaranteeing sponsorship of these projects (CF7). Furthermore, "a culture of data ensures that there is a focus on maintaining high data quality and accuracy, while a culture of customer listening that prioritizes gathering customer insights, actively listening to feedback, and incorporating the voice of the customer into decision-making processes places the customer at the center of strategic considerations" (CH16). Next, an initial multidisciplinary team is formed, bringing together all relevant stakeholders' perspectives and possible issues to ensure inclusive AI (CB3). This multidisciplinary team serves as the foundation for establishing an ethics committee in charge of investigating and framing AI in line with the ethical values upheld by the firm (EC21). According to the interviewee, the multidisciplinary team should include those with diverse competencies, such as a chief data officer, a domain expert with digital data and business insights, a data scientist, an IT professional, an ethics expert familiar with AI's legal and social implications, and a requirements engineer to translate ethical concerns into technical specifications (CJ22; EB20; PA8; C00; CB3).

Step 2: Ethics by design for customers, users, and society. Ethics by design refers to the process of considering ethical principles from the

beginning of the AI project, thereby adopting a proactive approach to ethics (Bourgais & Ibnouhsein, 2022). While insights from participants underscored the practical implementation of ethics in AI projects, extant literature on ethical AI practices also supports this step strongly. Ethics by design requires an ethics committee, i.e., managers and individuals with ethical expertise who are representatives from different regulatory bodies, but also domain experts, so that ethical considerations can be framed directly into the design of the AI application, rather than being viewed as optional (C00; EB20; CE6). At this stage, for each AI application in CRM, the ethics committee discusses and defines ethical principles and implications, as well as actions and tools, to align AI with the company's strategy (PD11; PE12; EB20; CE6). For example, when using natural language processing (NLP) to expedite customer mailing, ethical considerations arise regarding sentence construction and the potential for discrimination. Table A2 provides practical definitions of these seven ethical principles (Hub France IA, 2023) and examples related to CRM. The committee collaborates with executives to identify discrimination to avoid and to establish guidelines (EB20). For example, email texts should not perpetuate undesirable stereotypes, such as those tied to gender or ethnicity. This entails involving domain experts and various stakeholders in the discussion to incorporate diverse perspectives (CJ22; EC21; CE6; EA19) and foster collaboration and responsibility (CB2; CG13).

Step 3: Customer data comprehension. AI crossed with big data expands availability of customer information. Therefore, data capital comprehension is critical to understanding which data are available for the AI application, who manages the data, where they are, for what purpose they have been collected, and how they are managed (e.g., PA8; CI17; CD5; see Table A1 for full quotes). It is a time-consuming process, but fundamental to understanding whether the necessary data are useful for AI (data usefulness understanding), as interviewee CA1 explained: "There was all the analysis work, in which we said why do we have to take these data, rather than ignoring them." Respondent CD5 explained that the project's failure was caused precisely by the inability to understand that the available customer data were not enough to run the AI application successfully. Effective AI-CRM integrations emphasized analyzing data access thoroughly, asking, "What data do I have access to, and what performance can it achieve?" (C00). The team must assess legacy system constraints to ensure data integration, and legal teams must ensure GDPR compliance to avoid misuse of customer data (CG13; CH15). This structured approach helps prevent challenges related to data limitations and legal barriers. For example, "You must take care that the data is not sufficient to uniquely identify a person. Information such as age, email, sex, and address are anonymized and allow you to build the algorithm to identify, for example, the risk of customer abandonment" (CE6).

Step 4: Framing of AI-CRM organizational integration. During this step, the following elements emerged from the interviews: Outline how to achieve the defined goal through AI, whether to develop the AI solution internally or rely on external suppliers, estimated investment costs, the project's ownership, the project's sponsor, and the deadlines (see Table A1 for full quotes). Regarding the first element, one participant noted: "It takes a lot of time and is fundamental to transform the problem from a semantic level into something actionable, to convert intuitive and expert-based knowledge into a system that can automatically derive insights from data" (CG13).

Furthermore, to frame a model that remains effective in the future, it is crucial to analyze the external environment – including market trends, regulatory or technological changes, and socioeconomic disruptions (CI17; PE12; CG13) – "by engaging with field experts to identify any drastic changes in data concepts." For example, "the COVID-19 pandemic significantly altered customer relationships, rendering many models obsolete" (PE12).

Additionally, all members of the multidisciplinary team are delineated, along with their competencies, roles, and functions (e.g., CB3; PA8; PE12): "With the help of human resources, we engaged in an initiative to renew the teams, focusing on upskilling and reskilling, and we created a "One Team." This team would integrate both tech and specialized roles, including marketing, CRM, and e-commerce with different skills that need to be leveraged" (CH15).

The team, along with the ethics committee, operationalizes ethical principles, focusing on streamlined strategies such as compliance with regulations, such as GDPR (CJ2); privacy integration from the outset (PE12); and maintaining internal governance, even for outsourced AI tools (PA8). Including diverse stakeholders, such as data scientists and marketing managers, ensures a robust and ethical AI-CRM integration process (EB20). Decision matrices are used to maintain ethical principles throughout the AI-CRM integration process, and regular updates to ethical guidelines ensure alignment with emerging AI challenges (PE12; CJ21; CG13).

Afterward, companies define the AI-CRM integration process. First, they set the outputs and boundaries, which means "planning how to turn the people's know-how into data and algorithms, and how to bring out the results" (CG13) and identifying the processes on which the solution could impact. To do this, "you must have visibility over the whole process" (CE5). In addition, an analysis with the IT department on how to embed the model into business operations is paramount (CG13) to understanding whether there are constraints or blockages from legacy systems (C00).

Finally, participants outlined the importance of planning how to measure and read the results (CG14; PE12), and how to scale the AI application (PA8; CC4; PC10) (e.g., do we start with the first team and then extend to all teams, or do we want to deploy 100 % in the first run?; CJ22). As for the need to plan for measuring and interpreting results, one participant explained: "It is crucial to include AI experts to ensure that solutions stay intelligent over time. You need to incorporate the necessary modules to track the performance at the project's outset, not at the end" (PE12).

4.2. Macro-phase two: data governance and people learning and growth

Effective AI integration hinges on strong data governance and ongoing learning. Participants highlighted the critical need for upskilling internal teams, particularly data scientists and "translators" – individuals who can bridge the gap between technical teams (e.g., data scientists) and nontechnical stakeholders (e.g., business managers, marketers, and sales teams).

Step 5: Data governance throughout AI-CRM integration and evolution. Extant research has underscored that effective data models are crucial for efficient data organization and management in AI-CRM integration (Chatterjee et al., 2019). Interviewees provided valuable insights that enhanced our understanding of this process. Data modeling involves designing the conceptual representation of customer data and defining its structure, format, and rules for data storage, access, and manipulation (Ngai et al., 2009; Overgoor et al., 2019). Simultaneously, data creation addresses missing data, while data collection ensures only that relevant data are gathered from internal and external sources, ensuring efficient use and integration across structured and unstructured formats (e.g., social media posts, audio messages) (e.g., CE6; CG13; see Table A1 for full quotes). To minimize errors, interviewees emphasized the importance of avoiding manual data retrieval, replacing subjective inputs (such as conversation recall) with logs (e.g., emails, meetings, calls), and using structured fields (e.g., drop-down lists) that prevent information dispersion (PD11).

Then, data validation is performed to ensure data accuracy and quality, including data cleaning, and checking for duplicates (PD11). Regular meetings also should be held to validate essential ethical AI principles, including data privacy, fairness, and technology's societal or environmental impact (CH14; CH15; PE12).

Step 6: Customer data centralization throughout AI-CRM integration and evolution. Customer data centralization in AI-CRM integration involves three key components: customer data structuring; customer information

reorganization; and the effort to gather and consolidate all customer-related information into a single centralized system. This step requires labeling, synchronizing, and integrating the data with the CRM system and other data sets (e.g., ERP; e-commerce) (e.g., CG13; C00). Additionally, CRM processes can be redesigned (CG13; PE12; CA1), with data reorganized through a machine perspective and not simply mapped and automated from a human perspective (CE6). Companies then proceed with centralizing all the information around the customer into one system to provide a unique, complete, and omnichannel customer vision (e.g., CE6; CB3). Some participants called it a customer data platform, and, others, a data lake. Whatever they are called, both store vast amounts of structured and unstructured data, enabling a comprehensive customer view.

Step 7: People learning and growth throughout AI-CRM integration and evolution. To conduct learning and growth among employees during AI-CRM integration successfully, interviewees highlighted several key issues. First, it is essential to adopt a multidisciplinary and multicultural approach by involving managers from various departments, which enhances overall understanding (C00; CF7). Participants also pointed to the need to engage with training programs by recommending comprehensive and interactive sessions on AI for all employees that clearly explain its purpose, capabilities, limitations, and risks, thereby fostering trust in the technology (PC10; CG13). Additionally, maintaining ongoing communication and internal resource development is crucial to addressing employee concerns (CC4; CH15). Finally, incorporating validation of AI ethics principles – particularly accountability and well-being – into training discussions strengthens employees' ethical awareness and responsibility further (CG13; PE12; CJ22).

4.3. Macro-phase three: AI model implementation and governance

This phase covers development, validation, and deployment of the AI model within CRM, prioritizing transparency, user involvement, and ethical oversight. Unbiased data practices and strategic retraining ensure alignment with business goals and customer needs, fostering adaptive and accountable AI-CRM integration.

Step 8: Development of the AI model applied to CRM. The data are divided into sets for training, validation, and testing of the AI model (C00) to fine-tune it before reintroducing real data for accuracy verification (PD11). The training dataset identifies key impactful variables and eliminates those without effect, focusing on finding essential "ground truths" for effective system training (CI17; CG13). To ensure an unbiased model, it is crucial to remove discriminatory factors using manual reviews to detect and address skewness (CF7). AI model validation then employs the validation dataset to balance data accuracy and algorithm efficiency (C00; CG13; CF7), thereby identifying potential prediction distortions and ensuring adaptability to changing business scenarios.

Critical emphasis is placed on evaluating ethical explainability and transparency principles (CF7; PC10; C00). Incorporating explainability is particularly relevant for predictive AI models, as it often requires breaking down prediction generation and providing intermediary results (CG13). As an example of good practice, clarifying algorithm creation and the top three factors influencing predictions can enhance transparency (PC10). Despite efforts to achieve transparency, AI systems, particularly deep learning models and neural networks, face inherent complexities. These systems can embed biases that lead to prejudicial choices, often lacking clear visibility into their decision-making processes (CG13). The knowledge within these models can become inscrutable, complicating user understanding of how biases influence outcomes.

Step 9: Users' feedback analysis for AI model testing and adjustment. The final model is released into the test environment for AI model testing, in which balancing performance and ethical issues is key. This balance involves using data responsibly while ensuring personalization (CF7), adhering to ethical guidelines (CG13, CJ22), and maintaining transparency to build user trust (PE12). To manage these trade-offs successfully, organizations can integrate regular A/B testing (PC10; CG13), favor an end-to-end approach (CC4), avoid periods such as summer holidays when engagement might be low (CB3), and conduct user feedback analysis before fall rolls out (CF7; CB3; CG14). These practices help fine-tune the AI model in real-world settings, ensuring that it meets business goals while maintaining fairness and transparency.

Many participants stressed the importance of having humans on the loop to take corrective measures to adjust the model, thereby increasing the firm's accountability for it, checking biases, and achieving human autonomy and oversight (CF7). Good techniques include fallback to human operators (CG14, PF18), AI predictions and human judgment combination (CF7), satisfaction assessments (PC10), and regular monitoring of data and concept drift by experts (C00; CB3; PE12). Indeed, regular data updates necessitate periodic reviews to address any new biases or ethical concerns (C00; CG13; PC10; CJ22). This is achieved through continuous feedback loops, which revisit earlier stages, such as those related to data governance, to ensure that both the AI model and governance practices remain aligned and effective (C00; CF7; PE12).

Step 10: Rollout the AI model applied to CRM. Finally, companies can proceed with the rollout of the final model, running the real data set and putting it into production. Companies first define how they want to integrate the expected output in the users' workstation (CJ22). Once this is established, The AI model is integrated into the IT system, building data journeys and necessary connections while also addressing integration challenges (CJ22), particularly those related to incorporating the model into older legacy systems that often require specialized adjustments and careful planning (PC10). Collaborative planning between business and IT teams is essential for this phase (PC10; CG13). With an emphasis on effective change management and comprehensive user training (CF7; CJ22), the rollout can be optimized to ensure the AI model's success in production.

Step 11: AI model retraining and governance throughout AI-CRM integration and evolution. After rollout, participants highlighted that algorithm retraining is conducted periodically based on changes in inputs and outputs (CG13; CJ22; C00). Retraining frequency varies depending on the system – customer behavior monitoring might require updates every few months, while technical systems may need daily adjustments (CG13). Employees and end-users' feedback helps identify performance issues that data alone may miss (CF7; CJ22; C00). Data scientists are responsible for managing these adjustments and ensuring model reliability over time (CF7). Continuous monitoring detects drifts, and automated alerts can prompt retraining to maintain accuracy and alignment with business goals and ethical standards (PE12; PC10).

4.4. Macro-phase four: assessment and improvement of AI-CRM integration

Before scaling the project, it is essential to assess whether the AI integration meets performance targets, upholds the company's ethical standards, and ensures that users understand and trust the technology.

Step 12: Performance assessment throughout the AI-CRM evolution. For AI integration assessments, KPIs are measured and monitored for variations regularly (CG13). Maintaining alignment between KPIs and business objectives is a dynamic process that requires continuous evaluation and adjustment (CF7; CG13).

Human oversight remains essential in this process, providing critical context that AI systems may overlook. Experienced employees can validate and refine AI outputs, ensuring greater accuracy and relevance through their insights (CA1; CC4). To support this, data scientists actively monitor the system, ensuring that the AI adapts to new data and evolving business needs while maintaining the balance between automation and human judgment (CF7; CE6).

Step 13: Scaling of the AI application. Companies often start by applying AI to a single product or region, testing reliability before

expanding to a larger scope (PC10). The shift from short-term proofs of concept to long-term, sustainable AI models ensures scalability and integration into broader CRM strategies, focusing on "proof of permanence" (PA8). Customizing AI models for different markets and legal contexts is essential, as regulations and customer behaviors vary (CC4). Finally, collaboration between technical teams and other business units ensures that AI scaling aligns with business goals and maintains sustainability (CL24; CJ22).

5. Discussion

As summarized in Table 1, extant literature on AI integration steps has examined several key activities in AI integration into business, such as defining project goals, setting KPIs, data preparation (Herremans, 2021; Shrivastav, 2021), broad training (Lee et al., 2019; Makarius et al., 2020; Reim et al., 2020), and continuous monitoring (Holmlund et al., 2020). Our results delve deeper by structuring AI integration into the CRM context as coherent, interconnected steps from end to end. This structured approach is necessary given CRM's cross-functional and iterative nature, making AI integration particularly complex. Unlike general AI integration models that portray processes as linear (Fenwick et al., 2023; Herremans, 2021; Lee et al., 2019; Reim et al., 2020), our findings highlight that AI-CRM integration must be inherently iterative, frequently revisiting earlier phases – especially when handling complex customer data flows and the ongoing need for personalization and adaptation.

We identified several foundational elements and unique aspects that are essential for successful AI-CRM integration. The following sections elaborate on these elements, clarifying what is already known and how our study advances the field, detailing how CRM's specific features and dynamics impact the integration process (see Table C1 in Appendix C for a detailed summary).

Cultural foundations and strategic alignment. AI-CRM integration hinges on establishing both a strong cultural foundation and strategic alignment from the outset. Table 1 indicates that existing AI integration frameworks, such as Herremans (2021), typically suggest cultural shifts as a final step. In contrast, our research emphasizes the need to nurture a culture of data awareness and attentiveness to customer needs before initiating AI-CRM integration.

Project sponsorship. In addition, the literature highlighted project sponsorship's importance in AI adoption (Bonetti et al., 2023), and our findings align with this, but emphasized the dual nature of sponsorship further – incorporating both technical and ethical dimensions. This dual approach ensured that technical needs were met while maintaining transparency and fairness, particularly in handling sensitive customer data.

Early definition of KPIs and continuous performance monitoring. Other empirical evidence from our study is the early definition of KPIs and the continuous monitoring of AI performance throughout the implementation phase. This forward-thinking approach was vital due to the customer experience's centrality in CRM operations. While several studies in Table 1 emphasized the importance of defining KPIs and tracking (Herremans, 2021; Holmlund et al., 2020), they viewed performance monitoring as a late-stage activity. Instead, our framework integrated continuous reviews and active adjustment of AI models since the implementation phase, thereby ensuring that they remain aligned with CRM objectives. Moreover, ensuring reliability and ethical quality necessitates frequent reassessment, iterative adjustments, and close collaboration with end users – particularly in CRM environments that inherently demand iterative and personalized interactions (Payne & Frow, 2005).

Top management support and cross-functional collaboration. Our framework highlights the critical importance of cross-functional collaboration in AI-CRM integration, involving departments such as sales, marketing, IT, and compliance from the beginning. While Table 1 highlights that extant literature already has emphasized the need for cross-functional collaboration (Herremans, 2021; Holmlund et al., 2020), our study offers new insight by advocating for integration of cross-functional collaboration from the very beginning of the AI-CRM process. Indeed, embedding cross-functional teamwork early on is key to overcoming the siloed information structures typical of CRM environments that hinder communication and coordination (Malthouse et al., 2013; King & Burgess, 2008). Furthermore, given CRM's inherently data-driven and multidisciplinary nature, our findings suggest that strong support from top management is crucial to ensuring coordination and scalability across both technical and nontechnical teams, consistent with extant studies (Lee et al., 2019; Reim et al., 2020). In particular, AI applications in CRM impact multiple departments (e.g., marketing, sales, IT) and, therefore, require a top-down approach to ensure alignment with broader business goals, ethical standards, and legal requirements. The literature on AI frameworks divides management of AI projects into centralized or decentralized approaches (Herremans, 2021). Our study suggests that a centralized approach is more effective for AI integration into CRM, thereby ensuring better coordination across departments.

Ethics by design. While extant studies have recognized ethical concerns in AI adoption, they generally have not provided actionable guidelines on how to integrate ethical oversight into AI design and implementation (Jobin et al., 2019; Kumar et al., 2020; Zdravković et al., 2022). Unlike these general frameworks, our research explicitly incorporated ethics by design as a structured approach within AI-CRM integration. Our study highlights concrete mechanisms, such as defining ethical principles upfront, leveraging a dedicated ethics committee and involving cross-disciplinary stakeholders. Extant studies have suggested using ethics committees (Bourgais & Ibnouhsein, 2022), but our study elaborated on their specific role in CRM, focusing on sensitive customer data. We stressed that ethical oversight should be integral to AI design, with responsibilities such as establishing ethical guidelines tailored to CRM, defining ethical principles (see Table A2), and conducting regular compliance reviews throughout AI-CRM integration.

Strategic planning and scalability. Our findings highlight the critical importance of involving domain experts' early in the stage, ensuring effective communication, and considering external environmental factors during planning to facilitate smooth AI integration into CRM operations. While Herremans (2021) underscored the need to define AI project goals and involve domain experts, our study expands on this by showing that expert engagement should not be a one-time step, but rather an ongoing process. Unlike extant literature, which often treats communication as a late-stage activity (Lee et al., 2019; Reim et al., 2020; Makarius et al., 2020), we argue that communication should be initiated early in the process to align stakeholders and reduce uncertainties. Furthermore, we stressed the need to consider external environmental factors during planning, an aspect often overlooked in extant literature. Indeed, our findings indicate that many organizations engage field experts to monitor emerging trends, regulatory shifts, and other industry disruptions.

Finally, whereas extant research has considered scalability during the later stages of AI implementation (Holmlund et al., 2020; Shrivastav, 2021), our findings highlight the need for scalability planning from the framing phase. This early approach ensures that AI models are equipped to adapt to future growth in customer base, data volume, and market dynamics.

Data comprehension and centralization. Data preparation, evaluation of data availability and quality, and data collection and storage have been highlighted in extant studies as key determinants of AI integration (Herremans, 2021; Holmlund et al., 2020; Shrivastav, 2021; see Table 1 for an overview). Our study expands this understanding by emphasizing the importance of data comprehension and customer data centralization across all organizational touchpoints. Indeed, our findings went further by stressing the need to determine which data are actually necessary and accessible for AI-CRM integration. This underscores the significance of

fostering a culture of data sharing and implementing effective knowledge management practices within CRM environments (King & Burgess, 2008). These aspects are essential for protecting sensitive information and offering targeted decision support, especially in the face of fluctuating data access dynamics. Furthermore, our findings address the complexities of integrating diverse customer data sources into a unified, omnichannel view. This holistic, 360-degree perspective centralizes customer data onto a single platform, enabling consistent, actionable insights across all channels and organizational touchpoints. Such centralization goes beyond basic system integration, ensuring that CRM environments maintain data accuracy, regulatory compliance, and an effective understanding of customer needs, which are all essential for building and maintaining trust (Boulding et al., 2005).

User involvement and tacit knowledge. Our emphasis on user involvement from the planning phase sets our work apart. Table 1 indicates that extant studies on AI integration steps, such as Shrivastav (2021), typically have considered user feedback only during the later stages of AI model validation. The interviews revealed that much about users' knowledge required for successful AI-CRM integration is tacit, i.e., not easily documented or transferred. This aligns with De Bruyn et al. (2020), who mentioned the growing challenge of transferring tacit knowledge between AI systems and marketing stakeholders. To bridge this literature gap, we proposed some actions to facilitate the transfer of this knowledge from humans to machines: conducting face-to-face interviews with domain experts and users to clarify how to achieve goals through AI; gathering experiential knowledge from users through vocal instruments and translating them into analyzable data; extracting patterns from AI model training datasets to identify key variables; and involving humans in the loop, including users and customers, to validate AI model results interactively and ensure effectiveness even after human intervention.

Ethical feedback loops and adaptability. Table 1 indicates that extant literature on AI integration steps previously has suggested implementation of feedback loops for continuous improvement (Herremans, 2021; Holmlund et al., 2020; Reim et al., 2020). However, our framework builds on this idea by incorporating specific feedback loops and iterative assessment mechanisms to address ethical challenges, such as bias and fairness, more comprehensively. In developing AI models, particular emphasis is placed on balancing personalization goals with data privacy requirements – a nuanced and critical trade-off that is especially relevant in CRM contexts, in which personal data and customer relationships are at stake. By retraining and refining the AI model continuously, organizations can ensure that it remains adaptable and responsive to evolving customer behaviors, preferences, and market dynamics.

6. Conclusions

This study addresses the lack of thorough research and empirical evidence concerning integration of AI into CRM by providing valuable insights and guidance to practitioners and researchers in this field. Based on our research methodology, which employed maximum variation sampling (Coyne, 2018), our findings are generalizable across diverse organizational contexts and industries. By engaging companies that span different sectors – such as manufacturing, service, and retail – we ensured a comprehensive representation of real-world scenarios. Furthermore, thanks to a systematic approach that includes both within-case and cross-case analysis, we derived a unified framework applicable across various AI applications in CRM. Even though the participants implemented different AI solutions both in terms of AI type and functional area (marketing, sales, or after-sales), they shared a common approach to integration, demonstrating the framework's broad applicability.

6.1. Academic contributions

Our findings on AI-CRM integration extend the extant literature by offering a holistic, empirically driven perspective on how to implement AI in the CRM domain. Unlike extant studies, which tend to focus on isolated aspects of AI integration, our work provides a comprehensive set of steps, each addressing CRM systems' unique characteristics. We filled literature gaps left by extant research by delivering an operational guide that captures not only strategic decisions, but also practical activities essential for successful integration. In doing so, we responded to Loureiro et al.'s (2021) inquiry into how organizations should manage and implement AI systems, while also aligning with CRM strategies that pose specific challenges in AI integration (Mäntymäki et al., 2022).

Key findings indicate that successful AI integration hinges on some critical factors – such as top management support, sponsorship, data quality, security, and governance – already mentioned in the literature. However, our study extends the discourse by emphasizing how these elements interact specifically within CRM contexts, in which complexity management, data centralization, cross-functional collaboration, and ethical considerations are particularly salient. This distinguishes our work from generic AI models by providing a CRM-specific, step-by-step approach to addressing these unique challenges, going beyond merely avoiding failure factors. However, it is crucial to emphasize that while important, these elements alone do not ensure successful AI-CRM integration. A balanced and context-aware approach is essential for effective AI integration into CRM systems. As several participants in our study highlighted, overly ambitious objectives or misaligned expectations can be significant contributors to project failure.

But what distinguishes our work further is its empirical approach. While Herremans (2021), Holmlund et al. (2020), and Reim et al. (2020) developed conceptual studies or literature reviews, our findings are grounded in data and firmly anchored in real-world evidence, capturing different professionals' perspectives. Therefore, our study contributes to the broader academic discourse by showcasing the value of an empirical approach. Through in-depth interviews with industry professionals, we captured AI-CRM integration's nuanced realities, providing insights into the human and ethical dimensions that often receive less attention.

6.2. Managerial contributions

Our study provides insights for practitioners seeking to integrate AI within CRM systems based on comparative analysis of successful and unsuccessful implementations. First, strategic alignment and strong top management sponsorship appear to contribute positively to addressing cross-functional and ethical challenges. Establishing multidisciplinary teams and including ethics experts may support more responsible AI use. Second, fostering a culture of data awareness and attentiveness to customer needs – through attention to data quality, centralization, and continuous user engagement – can help align technical efforts with customer needs. These elements seem to play a critical role in supporting the sustainability of AI being integrated into CRM regardless of the project's scale or innovation level. Indeed, while agile or innovative projects may allow for more flexibility initially, our case studies suggest that as AI-CRM integration grows and aims for long-term success, attention to the aforementioned elements appears to be increasingly important for scalability. However, their relevance and applicability may vary depending on organizational context.

Third, planning for scalability and governance – including technical infrastructure, feedback loops, model retraining, and adaptability to evolving customer behaviors – seems to be associated with longer-term project sustainability.

Finally, we shed light on elements that may inhibit meaningful integration: ambitious goals; legacy IT systems with fragmented information; a focus on data governance before customer data capital comprehension; lack of business involvement; internal resource development; limited user involvement in decision-making; and issues with

Fig. 2. Guidelines for managers on the integration of AI applications in CRM.

information accuracy, entry, and sharing. To help executives navigate the complexities of AI integration into CRM further, we developed practical guidelines derived from a synthesis and re-elaboration of our findings. This guide encourages executives to ask themselves critical questions and consider essential recommendations when approaching the transformation (Fig. 2).

Our results offer valuable guidance for organizations seeking successful integration of AI into CRM, as confirmed by two consultants from Sugar CRM and Microsoft Dynamics (see Table B1). They endorsed the framework for its specificity, comprehensiveness, and adaptability, noting that the embedded loops support iterative integration and continuous alignment with changing business needs.

6.3. Limitations and future research directions

A key limitation of this study is the interviews' scope, which may not have fully captured the broader CRM landscape. Future research could expand the scope by including a broader range of industries and stakeholders, such as customer-facing employees and end-users, to capture a more comprehensive perspective. Although we evaluated our framework's usefulness with two consultants, further testing and empirical research are needed to assess its effectiveness across different contexts. Moreover, future research could examine how AI advancements, such as autonomous AI systems, influence the AI-CRM integration process and whether they alter our findings.

Further research areas include exploring ethics-by-design approaches in CRM, which our findings highlight as crucial for data transparency, accountability, customer trust, and employee engagement. Many companies are establishing ethical oversight, suggesting a need for deeper exploration of ethical issues in AI-CRM integration, beyond data privacy and bias, encompassing societal impact and organizational culture.

In addition, our discussions revealed certain actions for transferring tacit knowledge between AI systems and marketing stakeholders, highlighting the need for further research into strategies that enhance collaboration and communication between human users and AI across all stages of integration.

CRediT authorship contribution statement

Cristina Ledro: Writing – original draft, Writing – review & editing, Investigation. **Anna Nosella:** Writing – review & editing, Supervision. **Andrea Vinelli:** Writing – review & editing, Supervision. **Ilaria Dalla Pozza:** Writing – original draft, Investigation. **Thomas Souverain:** Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

Table A1
Quotes illustrating how each AI-CRM integration activity is implemented.

Activity	Recommendations and quotes
Step 1: Strategic goal setting and team formation	
Formulate clear business goals	Break down broad objectives into specific, realistic targets to improve integration success. Employ a top-down approach to align AI initiatives with business strategy and prioritize achievable goals. Clearly define business objectives before initiating AI projects to ensure alignment with desired outcomes and to avoid resource misallocation. - Quotes: "It is crucial to clearly define your key business goals and understand the issues and performance challenges you face" (CB2); "Be able to formulate your key business goals. Be very clear about what is going on in terms of problems, in terms of issues, in terms of performance" (PA8); "We started from the strategy. Instead of following a bottom-up approach, where you expect initiatives to emerge organically, we shifted to a top-down approach. This included the maturity analysis, impact analysis, and the matrix to select key priorities. Our goal is to ensure we work on topics that will deliver measurable value" (CF7); "When dealing with AI, the first step is to determine what you want to accomplish. It might sound simple, but many companies want AI without a clear idea of their end goals or deliverables" (CB2); "You need to define what you want to deliver, and it will totally define the data you collect, analyze, and so on. So, it is very important to know what you want to achieve and then get a strategy to achieve it" (PD11); "It is crucial that whatever the project delivers is utilized to create value for the company. I have seen many companies attempt to launch projects using customer data, only to end up with scores or models that were not used because they were not aligned with the company's strategic priorities" (PE12). "We are at a bit of a dead end in developing this AI (to automate the reading, comparison, and analysis of policies at a global level) because we think it is too complex. I believe our ambition is too challenging. We took a very complex process (underwriting) and hoped that AI could help us" (CC4)
Clarify the performance you want to achieve	Consider multiple perspectives, including algorithms, customers, organizational efficiency, processes, and users/employees. Define clear metrics for success and develop a strategy for how each metric will be measured. Ensure that the desired outcomes are practical and achievable. - Quotes: "The objectives of the projects are not well defined from the beginning... you lose time because if you do not do it from the beginning then you can develop something that might not be usable" (CJ22); "We need a machine which never makes any errors... and we had to stop this. What do we want to achieve? It is very difficult sometimes to have the business explain what the KPI of success is from the beginning" (CJ22); "We must always define with the business one metric on which to evaluate the performance of the AI and know it in advance before production" (CL24); "You need to have a clear idea of what you want to improve and why—define indicators, what to expect... For each advisor, it is necessary to compare the performance before and after the implementation of the system" (CM24); "Identify those KPIs at the beginning of the project to follow the performance of the solution, which companies often forget" (PE17)
Multidisciplinary team formation	Incorporate the viewpoints and potential concerns of all relevant stakeholders. Include IT specialists, data scientists, marketers, and domain experts to form a well-rounded team. Foster collaboration across all disciplines to ensure a holistic understanding and to mitigate biases. - Quotes: "Form a multidisciplinary team, including data scientists and key stakeholders from customer-facing roles... their feedback is essential throughout the project" (CB3); "Depending on the projects, the members of the multidisciplinary team may suffice to compose the ethics committee. They can also engage academic specialists or experts who provide complementary insights to better apprehend the variety of thought and cultural anchors of AI ethics, as well as the corresponding risks and breakthroughs in specific use cases" (EC21); "Involve a small group from business and IT to clarify the problem, define KPIs, and evaluate data needed for the project" (CJ22); "Integrate interdisciplinary skills... fostering new teamwork within the company to co-develop the algorithm and identify potential ethical biases"; "Involve experts who can provide diverse perspectives, even with the same moral principles" (EB20); "Companies are striving to create hybrid or multidisciplinary teams to help the business articulate its relevant needs" (PA8); "First of all, at a hypothetical discussion table, there must be a domain expert, data scientists, project managers, IT representatives, sponsors from senior management, and those focused on security and privacy" (C00); "Our team integrates data scientists with marketers and IT to align goals" (CB3)
Step 2: Ethics by design for customers, users, and society	
Establish an Ethics Committee	The committee should define ethical principles, identify potential biases, and create guidelines to avoid them. Involve diverse stakeholders for comprehensive oversight. Regularly assess ethical impacts on both internal practices and external outcomes. - Quotes: "Companies should have an internal ethics committee that adheres to broader rules, involving diverse representatives from different social components like sociologists, philosophers, and economists" (EB20); "There must be... those focused on security and privacy" (C00); "We have dedicated people with backgrounds in philosophy and data science to address cultural and ethical considerations" (CE6)
Discuss and define ethical principles	Adopt an "ethics by design" approach. Identify critical privacy and legal considerations, including customer data security, fairness, and non-discrimination. Assess how AI impacts individuals and communities at various levels. - Quotes: "Managing privacy by design was another challenge to face" (PD11); "Integrate privacy by design and adjust ethical concerns early in the project" (PE12); "Ethical checks should involve data scientists, managers, and individuals with ethical expertise, considering diverse perspectives" (EB20); "Implementing a privacy by design approach from the beginning requires ongoing interaction with privacy and security departments, ensuring appropriate data management and protection against threats" (CE6); "Set up guidelines and frameworks for ethical evaluations during the lifecycle of the AI model" (EB20)
Involve stakeholders	Engage domain experts to convert intuitive and experiential knowledge into data models and algorithms. Include users early in the process to improve model accuracy and ensure alignment with business needs. Align stakeholder expectations, especially when scaling AI solutions. - Quotes: "Aligning all stakeholders on the timeline and budget is challenging due to the uncertainty inherent in AI projects" (CJ22); "Stakeholders, including experts, regulatory bodies, and users, should be involved in defining moral norms that AI must respect" (EC21); "Broad visibility into the company's information assets helps protect against potential knowledge leakage" (CE6); "Stakeholders involved typically include data scientists, managers, consumers, and regulatory bodies to ensure ethical compliance" (EA19)
Be collaborative and responsible	Use collaborative brainstorming sessions to identify potential AI project opportunities, involving multiple stakeholders to ensure diverse perspectives are captured. - Quotes: "I have coordinated between various teams, including marketing, to gather insights from the public, and communication teams to design the emails we send to identified clients. We also collaborate with commercial lenders to help us define how to sell an insurance product and design a sales script. I provide all these models and work closely with various stakeholders in my role as product owner" (CB2); "It is important to sit down with domain experts because it is rare for artificial intelligence experts to

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Activity	Recommendations and quotes
	also be experts in the specific domain where the AI is being applied. In most cases, these two areas of expertise do not overlap" (CG13)
Step 3: Customer data comprehension	
Data capital comprehension	Clarify the scope and context of data available, aligning data collection and creation with specific business goals. Conduct feasibility studies to assess data adequacy for projects. Address challenges related to the original entry and quality of data. - Quotes: "What data are you collecting? Where is the data stored? Who is using it?" (PA8); "Define the output you want to deliver; it will determine the data you need to collect" (PD11); "Understand what kind of data you need to create, and for which business opportunity" (PE12); "We assess whether there is relevant data behind an initiative. It may not work due to insufficient data" (CL24); "We encountered difficulties in understanding how the data was originally entered" (CI17); "I would have enriched the dataset as much as possible. We only looked at navigation data and realized it wasn't sufficient" (CD5)
Data usefulness comprehension	Conduct in-depth analyses with data scientists to evaluate the usefulness of data points, ensuring each one contributes meaningfully. Revise data collection processes to close critical gaps and iteratively assess the relevance of collected data for AI applications in meeting business needs. - Quotes: "It is important to identify which data are useful and eliminate redundant or unnecessary information through thorough analysis" (CA1); "Assess how the insights derived from the data will address business issues" (PA8); "The first step is often to create data, but before doing that, you must understand what kind of data is needed" (PE12); "Critically evaluate what actions are being taken with the data already collected" (CC4); "Data quality and accessibility are crucial. People often cannot find data or do not understand the content, leading to delays" (CJ22); "Assessing data usefulness helps prevent the inclusion of redundant information" (CJ22); "Leverage underutilized data. Typically, only 10–20 % of available customer data is actually used" (CM25)
Data access comprehension	Identify and extract relevant data while eliminating redundant information to enhance system efficiency. Evaluate data quality and accessibility, ensuring relevance to business needs. Optimize data usage to minimize waste and address common issues of underutilized data in organizations. - Quotes: "What data do I have access to, and what performance can it achieve?" (C00); "Access and data quality are key issues. People often want to use data but either can't find it or do not understand the database structure" (CJ22); "Assess legacy systems early to plan for compatibility and migration" (CG13); "Ensure a unified strategy for data access across different systems and roles" (CH15); "Understand the limitations of legacy systems to avoid integration challenges" (CH15); "Ensure data cannot be used to uniquely identify individuals. Personal details like age, email, gender, and address should be anonymized to allow algorithms to work responsibly, for example, in assessing customer churn risk" (CE6)
Step 4: Framing of AI-CRM organizational integration	
Clarify how to achieve your goal through AI	Translate domain expertise into actionable, machine-readable formats by prioritizing key data inputs. Involve domain experts to ensure data processing and machine learning algorithms are aligned with contextual business needs, converting intuitive knowledge into structured insights. - Quotes: "Transforming a problem from a semantic to an actionable level is fundamental but time-consuming. The key challenge is converting expert intuition into a system that can automatically derive insights" (CG13); "The domain expert's role is crucial because the context may require a specific learning algorithm" (CA1); "Ideation starts with a business objective, such as using data to detect fraud" (CJ21)
Decide whether to make or buy the AI solution	Weigh the costs and benefits of in-house development versus outsourcing to determine the optimal strategy for AI solution development. - Quotes: "We manage development in-house, but bring in consultants when specific expertise is needed" (CB2); "Companies hire external agencies for large projects but keep strategic ones internal to retain control" (PD11); "We follow a 'make-or-buy' approach—outsourcing for fluctuating demands, while standard solutions are developed internally" (CJ22); "We currently manage everything in-house, though external solutions typically scale faster" (CH15)
Decide ownership, sponsor, people involved, and deadline ambitions	Establish clear roles and secure strong sponsorship for AI projects. Appoint data owners within each business unit to act as key points of contact, ensuring collaboration between business and data teams. Secure top management commitment to provide budgetary and strategic support for project completion. Assign project managers for coordination. - Quotes: "Top-level commitment is essential; without cohesion, the project can lose momentum" (CG14); "Strong support from leadership is necessary to secure funding and sponsorship" (PE12); "A project manager should coordinate the stakeholders and be supported by an executive sponsor" (C00); "We secured sponsorship at the C-level to ensure project success" (CF7); "During the framing phase, we determine the project sponsor" (CJ22); "You need a timeline, a responsible person, and a deadline" (CB3)
Analyze the external environment	Regularly monitor the external factors that may affect AI-CRM integration with the help of experts. Understand which regulatory, technological, or competitive shifts may impact the ongoing value of AI solutions. - Quotes: "The European directive sets key compliance requirements companies must follow" (EB20); "We analyze external trends that could impact AI performance" (CI17); "Regularly monitor regulatory shifts, involving experts to understand evolving concepts. For instance, the COVID-19 pandemic significantly changed customer behaviors, rendering some models obsolete" (PE12); "Anticipate technological developments that may affect AI systems" (CG13)
Build a multidisciplinary, multicultural, and multiskills team	Encourage collaboration between business and IT to prevent misalignment and bias. Embrace multicultural perspectives and diverse skill sets through initiatives such as "One Teams" and continuous reskilling. - Quotes: "Two things are essential: talented people with strong algorithmic skills and bridging skills—facilitators who connect technical experts with domain specialists" (CG13); "We engaged human resources to create a 'One Team,' integrating tech roles with marketing, CRM, and e-commerce" (CH15); "Hybrid teams help articulate business needs, focusing on how data can support goals" (PA8); "Data scientists worked closely with managers to co-create algorithms and understand potential ethical biases—multicultural teams helped train the algorithm". "Through interdisciplinarity of skills and multicultural teams, which also means acquiring new employees within the company, a company can co-build the algorithm and understand what ethical biases there can be" (EA19); "Multidisciplinary teams, including data scientists and customer-facing stakeholders, are key to ongoing success" (CB3); "Transferring project ownership to marketing – instead of the digital team - led to quadrupled adoption rates" (CJ22)
Plan how to ensure ethics principles	Ensure GDPR compliance, involve diverse experts to address biases, and balance ethical and practical considerations. Use decision matrices to maintain ethical principles and update guidelines as AI develops. - Quotes: "Marketing plays a frontline role in ensuring GDPR compliance through privacy-by-design principles" (CJ22); "Privacy by design is crucial; ethical concerns must be addressed from the outset" (PE12); "The CTO shifted focus to core principles such as data governance and customer privacy" (PA8); "Ethical evaluations should not be left to data scientists alone—managers and ethics experts are essential" (EB20); "We use decision matrices to identify and address key ethical issues before progressing" (PE12); "Ethical decision matrices help prioritize crucial issues during AI

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Activity	Recommendations and quotes
	development" (CJ22); "Ethical guidelines must be regularly updated to stay aligned with evolving AI challenges" (CG13)
Set output and boundaries	Establishing the outputs and boundaries of an AI project requires integration into existing ecosystems, including processes, IT components, and legacy systems. Involve all software management teams from the start to prevent obstacles during implementation. - Quotes: "Work closely with critical systems to ensure reliability and security—avoid over-reliance on external data lakes" (CG13); "Plan how to convert employee knowledge into usable data and algorithms" (CG13); "You must maintain visibility across the entire process" (CE5); "Integration must align with current processes and account for legacy system constraints" (C00)
Plan how to measure and read the results	Design feedback and performance evaluation mechanisms before implementation. Plan how to sustain AI effectiveness over time.- Quotes: "We designed the conversational agent to collect customer feedback at the end of interactions through open-ended surveys" (CG14); "Identify performance tracking indicators at the start to ensure long-term AI effectiveness" (PE12)
Plan how to embed the model in operations	Involve IT from the outset to ensure smooth integration with operational workflows. Define how AI outputs will be delivered and used by end-users, and clearly assign responsibilities. - Quotes: "We decided how many customers the claims team would contact weekly, based on AI output" (CJ22); "Engage IT early to integrate AI within the broader business ecosystem" (C00); "IT developed parallel data flows and system connections to enable smooth implementation" (CJ22); "Plan how data will be input and how results will be returned, ensuring contextual integration of the algorithm" (CG13); "Define how the AI model will be used in daily operations" (CG13); "Aligning AI results with operational workflows is key to achieving business impact" (CJ22)
Plan how to scale the AI application	Test the AI model on a small scale (e.g., by region or product line) before expanding it. Consider the specific requirements of different domains during the scaling process. - Quotes: "Once the AI model proves effective, the next major task is scaling it across the organization" (PA8); "We often begin with a single insurance product, like home insurance, before rolling out AI more broadly" (PC10); "Scaling across regions requires substantial customization" (CC4)
Step 5: Data governance throughout the AI-CRM integration and evolution	
Data modelling and creation	Develop a unified data model to align operations across different countries, ensuring consistency in how transactions are defined and recorded. Establish fields necessary for analysis, such as the number of employees, revenue, and industry classification codes. When data is unavailable, collect additional data, then prepare it for use by data scientists. - Quotes: "We focus on data modeling to align operations across all our countries and consolidate all sales data from different countries into a single database" (CF7); "Include fields related to the number of employees, revenue, and ATECO codes to identify the business sector" (CA1); "We gather all data and integrate it into a big data framework that includes unstructured data such as texts and images" (CE6); "Sometimes the data are available, but sometimes they are not, and we need additional data. Then IT has to prepare the data so that it is usable by data scientists" (CJ22)
Data collection	Systematically collect and centralize data using management tools, addressing gaps and adapting technology to capture overlooked data. Provide user-friendly tools and automate the process for comprehensive data collection and analysis. - Quotes: "The first part was data collection, and we used our business management system" (CA1); "Once we have the data, we store it in a data lake" (CJ22); "Take the data physically and put it in a place where we can analyze it" (CE6); "Technology now enables us to collect information that is useful today, even if it wasn't previously deemed necessary" (CG13); "Collect data from their systems and gather all sales activities to identify patterns and insights. We're currently rolling out a qualitative feature to capture important meeting details, such as whether it was a good meeting, the status of opportunities, comments on those opportunities, expected closure timelines, and next steps. To simplify the process, we've designed a user-friendly interface that prompts users to enter one to three next steps after a meeting. We're also implementing automatic connectors that pull in logs from emails, meetings, and calls. This will help us gather quantitative data and provide insights into the volume of interactions alongside qualitative feedback" (PD11)
Data validation	Ensure data accuracy and completeness through cross-verification, involving IT teams and data analysts. Establish validation checkpoints to confirm data quality before AI model training. - Quotes: "Mapping the system and ensuring everything is in order can take about one to three months. During this phase, we focus on verifying that all data is correctly placed, without duplicates, and aligned with the CRM" (PD11); "The challenge was understanding which data was actually useful and which were redundant" (CA1); "Regular validation steps ensure data quality and accuracy before integration" (CJ22)
Validation of the AI ethics principles of data privacy and regulation, fairness, and well-being	Conduct regular meetings to discuss personal data protection and access protocols, ensuring compliance with privacy regulations. Emphasize transparent data usage, obtain user consent, and create feedback loops for continuous improvement of AI models. Avoid including discriminatory variables (e.g., ethnicity) in training data and ensure equitable representation across user groups. - Quotes: "Topics concerning customer data are primarily managed by the Data Protection Officer (DPO), who organizes regular meetings - quarterly or bi-monthly - focused on personal data protection" (CH15); "We have a committee dedicated to data protection that meets regularly" (CH14); "Ensure that you are not using data without user consent and be transparent about how the data will be utilized. Prioritizing ethical data practices is essential" (PE12); "No algorithm should include ethnicity as a training feature. Ensure fairness of representation" (PE12); "Data should be anonymized or pseudonymized to protect privacy"; "Bias can stem from the data itself or from how it is used. Segment users into groups to avoid over-reliance on personal data" (PE12)
Step 6: Customer data centralization throughout the AI-CRM integration and evolution	
Customer data structuring	Label data to transform subjective insights into objective datasets, addressing challenges such as user-based labeling and ethical concerns. Establish governance and build structured platforms to ensure data quality. - Quotes: "Transform subjective human thoughts into objective data - a process called 'datafication'" (CG13); "Days dedicated to the evolution and structuring of CRM lead to cleaner results" (C00); "Mechanical labeling of messages functions as the training set for AI" (CA1); "Build a data platform and establish data governance" (CF7); "Mapping CRM data ensures accurate placement, synchronization, and elimination of duplicates" (PD11)
Customer information reorganization	Reorganize customer information to improve CRM efficiency by centralizing previously fragmented data and enhancing cross-channel communication. This includes integrating feedback and respecting privacy requirements. - Quotes: "We better structured and centralized customer information, incorporating cross-channel interactions" (CG13); "Personas evolve, leading to new ways of organizing data by relevance" (PE12); "Significant modifications were made to CRM's organizational structure" (CA1); "The process must no longer be done manually but automated. To do this, we need to collaborate with those who understand the process, as it should be redesigned" (CE6)

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Activity	Recommendations and quotes
Gather and consolidate all customer-related information into a single centralized system	Establish a centralized data infrastructure, such as a data lake, to integrate structured and unstructured data and provide a comprehensive, 360-degree view of customers. Centralizing data also facilitates effective algorithm development and testing. - Quotes: "We gather all data and integrate it into a big data framework, serving as a third level of abstraction - hence the term 'data lake.' Our data lake includes both structured and unstructured sources, such as text and images" (CE6); "We have designed a Big Data Lake containing data from banks as well as from other subsidiaries" (CB3); "We built a data platform to centralize all data. Now we can execute all our analytics use cases" (CF7); "Keep all data centralized at the point of consumption for a comprehensive, 360-degree view of the customer" (CD5); "All customer information should be unified in one system for a 360-degree view" (PC10); "Develop a better 360-degree view of our customers" (CJ22)
Step 7: People learning and growth throughout the AI-CRM integration and evolution	
Training of all people to stimulate trust and awareness around AI	Educate all users about the value of AI, engage stakeholders to build trust, and offer role-specific training, including on ethical implications. Continuously develop internal capabilities to ensure effective AI adoption. - Quotes: "Focus on explaining the value and impact of predictive models to end users" (C00); "Train all 300,000 employees, including those in the field, on data-related topics" (CF7); "We explain how algorithms are made to ensure ethical, responsible AI" (PC10); "First explain what artificial intelligence is and its limitations, then work collaboratively. Ensure users understand when they are interacting with a machine. Represent and address all potential scenarios fairly to avoid data bias" (CG13); "Ongoing internal resource development is needed to use AI effectively, but we never did this - this was one of the pitfalls" (CC4); "Training is prioritized across the organization, from the CEO to developers" (CH15)
Validation of the AI ethics principles of accountability and well-being	Establish accountability frameworks where stakeholders take responsibility for ethical compliance across the AI lifecycle. Regularly assess the social and psychological impacts of AI on users. - Quotes: "Regular assessments by an ethics committee ensure accountability at every phase" (CG13); "Ethical checks should not be conducted solely by the data scientists who developed the algorithm" (EB20); "Validation should include assessments of social and psychological impacts" (PE12); "We set up accountability measures to ensure compliance during AI implementation" (CJ22)
Step 8: Develop the AI model applied to CRM	
AI model training	Identify ground truths from collected data to improve model accuracy and engage domain experts to determine the most representative data. Remove bias by eliminating discriminatory elements and testing skewness. - Quotes: "There is a clear difference between the data used for training and the data that must be predicted over time. Training phase uses completed actions to help AI learn" (C00); "Set aside real data to validate the model after initial training" (PD11); "The learning phase needs a historical dataset to build an adaptive algorithm" (CI17); "Find 'ground truths' in the data to train and help the system learn correctly" (CG13); "Remove discriminatory elements to ensure training data is not biased" (CF7)
Validation of the AI model	Validate AI models using precision, recall, and balanced performance across outcomes. Run benchmark tests to ensure the model aligns with performance metrics. - Quotes: "During the model construction phase, we validated it by measuring its performance. We examined how the model performs concerning the possibilities of the classes to be predicted" (C00); "To effectively monitor the algorithm's performance, it is crucial to maintain control over the KPIs" (CG13); "Run regular benchmark tests to ensure performance is aligned with goals" (CF7)
Validation of the AI ethics principles of explainability and transparency	Make AI decision-making understandable through intermediary steps and factor explanations to enhance user trust. - Quotes: "The next step is to incorporate explainability into the algorithm by providing intermediary results. Transparency in AI is about breaking down decision-making into understandable steps" (CF7); "We explain all the algorithm steps and why a recommendation was made to ensure transparency" (PC10); "Consider what levers to disclose to avoid distorting user actions" (C00); "Explainability is essential to enhance user trust in the AI model" (CG13); "Explain the top three factors leading to each prediction for auditing and bias prevention" (PC10); "Biases in AI systems can be implicitly embedded, making transparency challenging" (CG13); "Transparency requires collaboration between data scientists and managers" (EB20)
Step 9: Users feedback analysis for AI model design, testing and adjustment	
AI model testing	Conduct thorough testing using pilot phases, A-B testing, and end-to-end evaluation. Ensure the timing of tests aligns with optimal business periods to avoid performance issues. Compare groups with and without the AI tool to verify its benefits and improve adoption. Avoid deploying the model without complete end-to-end testing to prevent overlooked issues during production. - Quotes: "Lead pilot phases to refine the algorithm" (CB3); "Conduct A-B testing to verify benefits, comparing agents with and without the tool" (PC10); "For instance, customer satisfaction and control group comparison are key for evaluating offer performance" (CG13); "If you do not test it end-to-end, you might find that something was overlooked in production" (CC4); "Avoid critical periods like summer for launching pilots" (CB3)
Achieve a tradeoff between performance and ethical issues	Balance personalization and privacy. Respect ethical guidelines and address customer concerns over data use. Quotes: "It is the paradox of customers: they do not want their data to be used, but they want personalized experiences" (CF7); "We have aligned the privacy component and ethical evaluation, following the guidelines issued by the European Union" (CG13); "It is difficult to make sure that discrimination is not active... GDPR restricts collecting data on religion or ethnicity" (CJ22); "Customers are more aware of the ethical use of AI and may prioritize trust over one-to-one personalization" (PE12)
Users' feedback analysis before roll-out	Engage users early in providing feedback to refine the model and ensure that it meets user expectations. Use user feedback to make necessary adjustments before the final deployment of the AI model to increase its acceptance and usability. Conduct iterative testing sessions to incorporate feedback into the development process. - Quotes: "By engaging in discussions with users, we often hold meetings during the algorithm development process" (CF7); "We generate results in Excel so they can review and interact with the data, providing feedback along the way" (CF7); "It is important to involve stakeholders early on because their feedback is essential throughout the project" (CB3); "Users' feedback analysis before roll-out ensures the AI model's usability" (CG14)
Humans on the loop to take corrective actions	Implement feedback systems and human oversight to enhance AI interactions. When AI systems falter, they escalate to human operators. Use metrics like net promoter score (NPS) to measure satisfaction and guide improvements, with human representatives addressing customer issues directly. - Quotes: "In instances where the conversational agent was not sufficiently effective, dialogues required intervention from a live operator" (CG14); "NPS measures user satisfaction, helping to identify improvements" (PC10); "Virtual assistant escalates contact to a customer service agent when uncertain" (PF18); "Claims handlers reach out to customers with high dissatisfaction scores to resolve issues" (CF7)
Data and concept drift monitoring	Adapt models to incorporate new information, account for product or behavior changes, and mitigate human biases that may impact the system's predictions. Engage domain experts to reassess the data context and ensure the solution

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Activity	Recommendations and quotes
	remains effective over time. - Quotes: "Regularly monitor the model for issues like data drift and other phenomena" (C00); "Model divergence can occur due to human interaction, requiring adaptation" (C00); "Database changes led to the need to rethink the algorithm for consistent performance" (CB3); "Ensure that your models stay smart by analyzing data drift and concept drift" (PE12)
Validation of AI ethics principles of fairness (prevent discrimination	Ensure AI model transparency to prevent misuse and address ethical concerns, such as bias and discrimination. Regularly evaluate data inputs and outputs to detect and mitigate ethical risks, ensuring fair and unbiased outcomes. - Quotes: "They need to understand which levers they can activate... taking into account distortions, including psychological ones" (C00); "The risk with AI systems is that these biases become embedded invisibly within the system... unlike in a textbook" (CG13); "The real risk is that you might amplify existing biases... ensure the recommendations produced are not biased" (PC10); "When using AI algorithms, if you do not set clear boundaries, you might unintentionally exclude these groups... it is challenging to ensure that indirect discrimination isn't happening" (CJ22)
Step 10: Roll out of the AI model applied to CRM	
Integrate the AI model within the IT system	Integrate the AI model within the IT system to ensure it functions smoothly with existing infrastructure. This involves collaborating with business teams to determine integration specifics, developing data connections, and training users on the new tool. Address challenges of legacy systems and finalize deployment logistics, including scope and user involvement. - Quotes: "First align with the business on how to integrate the data product into the workflow of the end users. For example, we decided how many customers would be called by the claims management team every week, ensuring a systematic roll-out. At the same time, IT development runs in parallel. Once we have a clear deployment plan, the IT team works on building data pipelines, system connections, and all necessary integrations" (CJ22); "The industrialization means integration within the IT system, and that costs money" (CJ22); "Integrating our system into their existing IT infrastructure is not straightforward, which raises concerns" (PC10). "You need to work closely with IT teams to ensure smooth integration" (CG13)
Roll out of the final model	Roll out the final AI model once the model (prototype) yields positive results and demonstrates value in real-world settings. - Quotes: "The prototype yielded positive results, leading us to enter the go/no-go phase. After observing the value and effectiveness of this approach in the field, we decided to industrialize the process and move it into full production" "We implemented a structured roll-out plan involving key teams to ensure smooth deployment" (CF7); "In parallel, change management begins. This involves training users, explaining how the new data tool works, and guiding them on how to use it effectively. We also strategize the rollout: Do we start with a pilot team and then gradually expand, or do we deploy it across all teams at once? Change management is a key part of the project, ensuring smooth adoption" (CJ22)
Step 11: AI model retraining and governance throughout the AI-CRM integration and evolution	
AI Model retraining	Retrain AI models periodically to adapt to changing contexts and maintain accuracy. The frequency of retraining depends on the nature of the data and the system's dynamics. It is considered the best practice to retrain models at least biannually, while some more dynamic systems may need more frequent adjustments, even daily. - Quotes: "Retrain the algorithm regularly, depending on how dynamic the context is" (CJ22); "Algorithms need to be periodically retrained based on real-world outcomes" (CG13); "Retrain at least twice a year" (C00); "You need to decide how frequently to retrain the algorithm. We retrain the one for alarms daily, but for customer data, every three months" (CG13)
Users' feedback analysis after roll-out	Incorporate user feedback post-rollout to validate AI model performance and adjust based on real-world input. Engage users, management, and commercial teams to identify discrepancies, ensuring alignment with expectations. Set up a feedback system and involve technical experts to maintain accuracy. - Quotes: "To test our algorithm, most of the time we work with employees to know if they are performant or not. It is worse to not ask them". "The idea now is to add the same feedback loop for algorithms. When users think the results are drifting or skewing, they need to request technical support from data scientists" (CF7); "Management, commercial teams, and end users were involved in the validation phase after the release" (CJ22); "For instance, the sales director says: let's take a look at your opportunities. This one is predicted to be a loss, let's go check it out" (C00)
Monitoring and Correction of AI model, its inputs, and outputs	Continuously monitor AI models with dedicated data scientists, using automated alerts to detect drifts. Provide user support through a helpdesk and retrain models as needed. - Quotes: "We have dedicated data scientists focused on maintaining the algorithms over time" (CF7); "Developing an internal helpdesk…users ask technical support to the data scientists" (CF7); "Monitoring algorithm drifts is essential. When we detect a drift, we issue a warning and retrain the model" (PE12); "Set up alerts to monitor and correct deviations in AI model performance" (PC10)
Step 12: Performance assessment throughout the AI-CRM evolution	
Measure and monitor key performance indicators	Maintain control over key performance indicators (KPIs) to assess the effectiveness of algorithms and whether they are meeting the performance targets set. - Quotes: "To effectively monitor the algorithm's performance, it is crucial to maintain control over the KPIs assigned to the system" (CG13); "Continuous performance monitoring and alert systems for deviations from KPIs" (CG13); "You must identify those KPIs at the beginning of the project to follow the performance of the solution" (PE17)
Check and adjustment of the AI model and its performance assessment	Human oversight ensures accuracy, adds valuable judgment, and prevents algorithm drift over time. Corrective actions are performed based on experience, and data scientists also maintain algorithms to ensure consistent performance. - Quotes: "Corrective actions to adjust the priorities assigned by the AI system, based on human experience" (CA1); "Someone must validate these results; AI may be good, but human judgment is often required" (CC4); "Data scientists dedicated to checking that algorithms are not drifting over time" (CF7); "There is always still some level of human interaction involved" (CE6)
Step 13: Scaling of the AI application	
Scale of the entire portfolio, business, or all geographies	Adapt to local legal and policy differences, avoiding a one-size-fits-all approach. Start with small implementations (like a specific product or region) before broader expansion. Focus on transitioning from proof-of-concept to full operational scale are key. - Quotes: "Companies often begin by implementing AI tools on one specific product before expanding to cover the full product portfolio" (PC10); "Starting in one region and then rolling out to more ensures robustness before a broader application" (PC10); "Executives are primarily focused on moving from isolated tests or laboratory models to integrating AI across all business operations or geographies" (PA8); "Applying AI models to other regions requires significant customization due to differences in policy structure… A one-size-fits-all approach does not work… Conduct a feasibility analysis before scaling" (CC4); "Feedback from projects that have been successful within the group is crucial" (CL24); "We developed a fraud detection system for health insurance and then promoted it to other lines of business, suggesting that they could also apply this solution" (CJ22)

Table A2
7 ethical principles for AI integrations in CRM.

Ethical Principle	Description
Privacy	No gathering or processing of data should occur without the user's informed consent. Legally speaking (e.g., GDPR 2016), features that are direct (e.g., "race" in the US) or indirect predictors (i.e., proxies), such as phenotype, religion, sexual orientation, or gender (e.g., the "postcode" potentially indicating ethnicity in the US, depending on data and the history of segregation), must not be used for predictions. However, privacy is not solely satisfied by avoiding illegal features for predictions. For example, the US platform Retail used a consumer's online transaction history to make recommendations. A woman who had just taken a positive pregnancy test bought baby products in advance, leading the platform to send her postal discounts on the topic. However, this inadvertently informed her entire social circle of her pregnancy, even though she did not wish to disclose it in the early stages. Considering edge cases where CRM systems might infringe upon consumers' privacy is a crucial aspect of privacy management. For us, privacy involves ensuring "that you are not taking data without consent, that you do not use too much data, that you are not choosing data you are not supposed to use" (PE12), and that data are "compliant with regulations" (PC10), while also "balancing consumers' rights/benefits and the data used" (PA8)
Safety	The AI used should ensure that consumers' data and communications are protected and preserved to provide stable results. Even if not legally required, information in e-commerce may need to be highly secure for the sake of consumers (e.g., the intimacy of care products recommended by a doctor, especially if the client does not wish to inform their relatives or friends). Given the "intimate" nature of transactions and communications, which is highly context-dependent, it seems to be the firm's duty to ensure there is no leakage of such data. This means implementing safeguards against hacking during the data collection process, as well as during the training and storage phases. Safety is not only a matter of data security: the stability of AI results must be ensured despite the inherent unpredictability of AI systems. This refers to algorithms whose functioning is not entirely determined by explicit rules written by the programmer, such as artificial neural networks, which are commonly used for target recommendation and upselling. This can be accomplished, for example, during AI monitoring (just after training and before deployment), by testing whether a model trained with similar individuals will produce similar recommendations
Fairness	To simplify, we approximate fairness with equality of treatment for group fairness (Hedden, 2021): individuals who share the same characteristics but differ only by a protected attribute (e.g., sex) should be treated similarly. More broadly, fairness implies the inclusion of diverse consumer types, who should not be rejected when desiring to access a product. Existing regulations in the EU, US, Canada, and other regions already forbid the use of proxy features, such as religion, for predictions, as these are considered discriminatory. More generally, fairness requires that, throughout its lifecycle, AI does not propose different content based on discriminatory traits. For example, it is crucial for CRM systems dealing with wealth to design a user interface that is simple, sparse, and visually clear for older users, ensuring they do not feel excluded. Fairness can be achieved by "ensuring that none of the training data are discriminatory or biased" (CH15), "guaranteeing a fair representation, meaning that all possible cases/scenarios must be considered and treated fairly" (CG13), and "avoiding the over-protection of certain groups of buyers/customers," i.e., ensuring minorities are not excluded (PC10). This can be achieved first by removing discriminatory variables during model training, and then by examining the results for potential biases (CF7)
Responsibility or accountability	Responsibility refers to the firm's ability to account for all decisions made in the design and deployment of the CRM tool, including the algorithmic components, the targeted audience, and all foreseeable consequences of the AI deployment. For instance, in the case of an online platform providing access to children's media, accountability involves considering the business model before launching advertisements. If the company allows any content without moderation, it will be held legally and morally responsible for this platform design, as it could be seen as conflicting with users' values. In contrast, forbidding certain types of content in the name of human values can have a positive impact on the firm's image
Explainability	Explainability involves providing information on the technical functioning of AI (transparency) and adapting to users' categories to make AI operations accessible (intelligibility). As noted earlier (see the 'Safety' principle and Table 2), AI is often based on complex algorithms that make it a "black box." For us, transparency involves a duty for designers to make as clear as possible the types of models used and the reasons for their selection, while striving towards models that are simpler to interpret for the CRM users. Furthermore, "transparency is a matter of explicability of the algorithm. Users should be informed when a system uses AI, whether they can disable or limit personalization, and when faced with automated decision-making technology" (PA8). In addition to the technical aspect, intelligibility requires aligning AI functionality with the user experience. For example, in the case of a cross-sell recommendation by the CRM, providing a natural language explanation of how the model reached its decision may help the CRM justify the action to consumers
Well-being or sustainable development	It is not enough for humans to merely "exist"; we ought to thrive. This means that our duties towards other humans, nature, and future generations should be integrated into the technical solutions we propose today. Among other considerations, care for the environment can lead to efforts to measure the energy consumption involved in data collection, storage, and model training - e.g., when recommending thousands of products on an evolving online platform. Designers can also seek ways to compute the carbon footprint of different models before selecting one, opting for the most energy-efficient (e.g., training data without storing it on terabytes of servers, which can also enhance privacy protection). Moreover, sharing information on energy consumption with consumers can enhance the company's image. Furthermore, since AI leads to rapid transformations in jobs and society, long-term social issues should not be overlooked, such as the outsourcing of low-wage labor for labeling services: "You must consider the type of services you are outsourcing during your project, and unfortunately, this is a very common issue nowadays" (PE12)
Autonomy	Derived from the Ancient Greek, meaning "to give oneself one's own law," autonomy calls for the integration of fundamental rights, human agency, and human oversight into the products we sell. While income benefits may sometimes suggest otherwise, certain applications of AI can act contrary to an ideal societal framework. For instance, in China since 2019, the "social credit system" has used facial recognition and data analysis to attribute liability scores to citizens, restricting those with insufficient "points" from using public transport or even visiting family. Although this might initially seem out of scope for CRM, the direction taken by use cases is never neutral. For example, the use of sentiment analysis to detect consumers' feelings before they are fully aware of their desires, or the application of spelling mistake detection, could inadvertently categorize less-educated clients. Therefore, firms should carefully inspect and validate the potential societal implications of each use case

Table A3
KPIs categories in AI integrations in CRM.

KPIs category	Description
Algorithm	Evaluate the technical performance and quality of the algorithm. For instance, "regarding virtual assistants such as assessing how much the person you talk to understands what you say is a fundamental thing, then evaluate how well the automatic system understands what the customer has asked and then look at the final evaluation that gave you the final customer" (CG13)
Customers	Evaluate the effect the model produces on the customers. For example, "the increased number of posts, conversion rate and then retention as well" (PC10); "enhanced quality for customer satisfaction" (CG13)
Organization	Evaluate financial, environmental, and social savings. For instance, company E "have made a retention list for clients with abandonment risk (predicted via AI)" and "analyzed how many of the ones they predicted went out compared to the normal customer base" and how much they "earned from those who should have gone out, but who withheld thanks to the discount campaign" (CE6)

(*continued on next page*)

Table A3 (*continued*)

KPIs category	Description
Processes	Assess quality improvement. Quality can be precisely "the timing, for example, the less things are broken, the faster they were repaired" (CG13)
Users/ employees	Assess the user'/employee's perception and satisfaction. For instance, "The Net Promoter score helps understand whether the users are satisfied with the tool, whether the suggestion they have for improvements" (PC10)

Appendix B

Table B1
Framework usefulness: some quotes from the interviews with two consultants.

Sugar CRM consultant	Microsoft Dynamics consultant
Utility: "I find myself in it. Then here you have gone deeper even with the questions". "For someone approaching it (the AI integration in CRM) for the first time and saying, 'Okay, I do not know how to do this, I do not know anything,' following it to the letter, maybe some steps are more comfortable just before or just after. So maybe explaining that the steps do not necessarily have to be followed in order; we can first take the framework, study it, get an idea of how to organize the work, and then start."	Utility: "It would be very useful, to tell the truth, because objectively when you do a CRM project the artificial intelligence component could be very useful for all the project phases". "It seems suitable to me for all types of companies. From what I have seen, it seems valid. Now, there is a high demand and what you showed me applies to any type of company."
Willingness to use: "I would have used it because it is very specific and therefore it is super useful because it covers all the points. Perhaps I would have been careful not to follow it in order."	Willingness to use: "I found it very interesting in itself and it would be very interesting to apply it effectively to a CRM project". "The critical issue I might find is that every situation has its specifics. However, I think it is adaptable because, naturally, it is a base that needs to be tailored to each specific situation."

Appendix C

Table C1
Key aspects of AI-CRM integration: Literature vs. empirical insight.

Key emerged aspects	Literature based evidence	Empirical evidence: new insights	Rationale
Cultural foundations and strategic alignment	Addressing culture late in AI integration (Herremans, 2021)	Cultural alignment from the start	CRM requires an early customer-focused culture to ensure alignment with AI integration
Project sponsorship	Importance of sponsorship (Bonetti et al., 2023)	Dual approach - technical and ethical sponsorship	Ethical concerns in CRM demand sponsorship that balance technical and ethical needs
KPIs and continuous performance monitoring	Performance evaluation post-deployment (Herremans, 2021; Holmlund et al., 2020)	Early definition of KPIs and continuous monitoring during implementation	CRM requires ongoing alignment with customer goals
Top management support and cross-functional collaboration	Need for top management support and cross-functional collaboration (Lee et al., 2019; Herremans, 2021; Holmlund et al., 2020; Reim et al., 2020; Shrivastav, 2021)	Cross-functional collaboration from the start and a centralized management approach	Inherently data-driven and multidisciplinary nature of CRM
Ethics by Design	General ethics discussion (Jobin et al., 2019)	Clear "ethics by design" framework, including ethical principles and ongoing reviews	CRM's sensitive customer data requires ethics integrated from the start to ensure fairness and transparency
Domain experts' involvement	Involve domain and AI expertise in AI project goals and problem statements (Herremans, 2021)	Builds on literature but highlights the early-stage, ongoing involvement of experts through the AI-CRM lifecycle	Ensure AI solutions are rooted in organization needs
Effective communication	Communication post-deployment (Lee et al. 2019; Makarius et al. 2020; Reim et al. 2020)	Need for early-stage communication, involving all stakeholders	Build trust, reduce resistance, and align stakeholders early on to minimize misunderstandings
Consider external environmental factors during planning	Not explicitly addressed	Importance of involving field experts to monitor external trends, regulatory shifts, and disruptions to ensure adaptive integration	Ensure that AI-CRM integration stays flexible and adaptable to changing market conditions
Scalability	Scalability considered later (Holmlund et al., 2020; Shrivastav, 2021)	Planning for scalability from the start	CRM's dynamic nature demands adaptability to changes
Data comprehension and centralization	Emphasis on data quality, preparation, collection and storage (Herremans, 2021; Holmlund et al., 2020; Shrivastav, 2021)	Focus on data centralization and customer data comprehension	Centralizing and understanding customer data are essential for building trust, ensuring accuracy, preventing integration issues, and maintaining regulatory compliance
User involvement and tacit knowledge	User feedback is often collected later in AI integration (Shrivastav, 2021)	Involve users early, especially to transfer tacit knowledge	Early user input is critical for aligning AI with CRM-specific needs and workflows
Ethical feedback loops and adaptability	Feedback loops for continuous improvement (Herremans, 2021; Holmlund et al., 2020; Reim et al., 2020)	Expands loops to address ethics, privacy, and fairness	CRM requires continuous ethical evaluation due to the sensitive nature of customer data

Data availability

Data will be made available on request.

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